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Hsueh et al.(10) **Pub. No.: US 2016/0259147 A1**(43) **Pub. Date: Sep. 8, 2016**(54) **IMAGING SYSTEM, IMAGE CAPTURING
APPARATUS AND ELECTRONIC DEVICE**(52) **U.S. Cl.**CPC **G02B 13/0045** (2013.01); **G02B 9/60**
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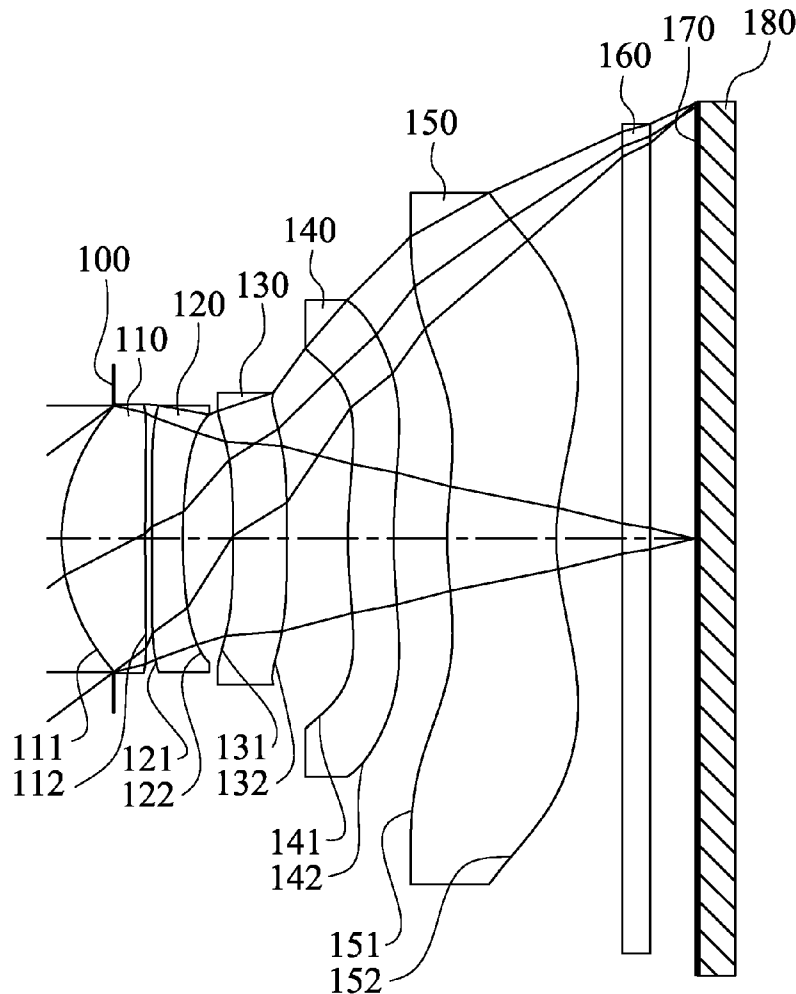
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ABSTRACT(21) Appl. No.: **14/727,302**(22) Filed: **Jun. 1, 2015**(30) **Foreign Application Priority Data**

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An imaging system includes, in order from an object side to an image side, a first lens element, a second lens element, a third lens element, a fourth lens element and a fifth lens element. The first lens element with positive refractive power has an object-side surface being convex in a paraxial region thereof. The second lens element with negative refractive power has an image-side surface being concave in a paraxial region thereof. The third lens element has refractive power. The fourth lens element with refractive power has an image-side surface being concave in a paraxial region thereof. The fifth lens element with refractive power has an object-side surface being convex in a paraxial region thereof and an image-side surface being concave in a paraxial region thereof.



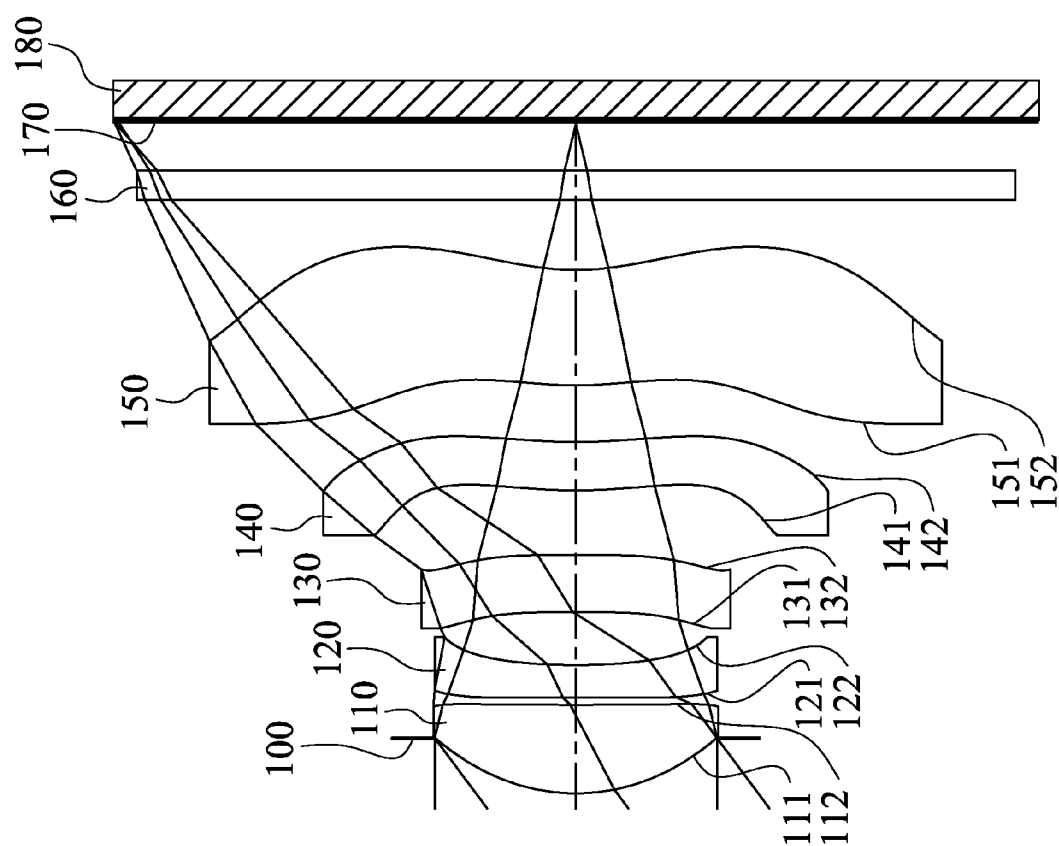


Fig. 1

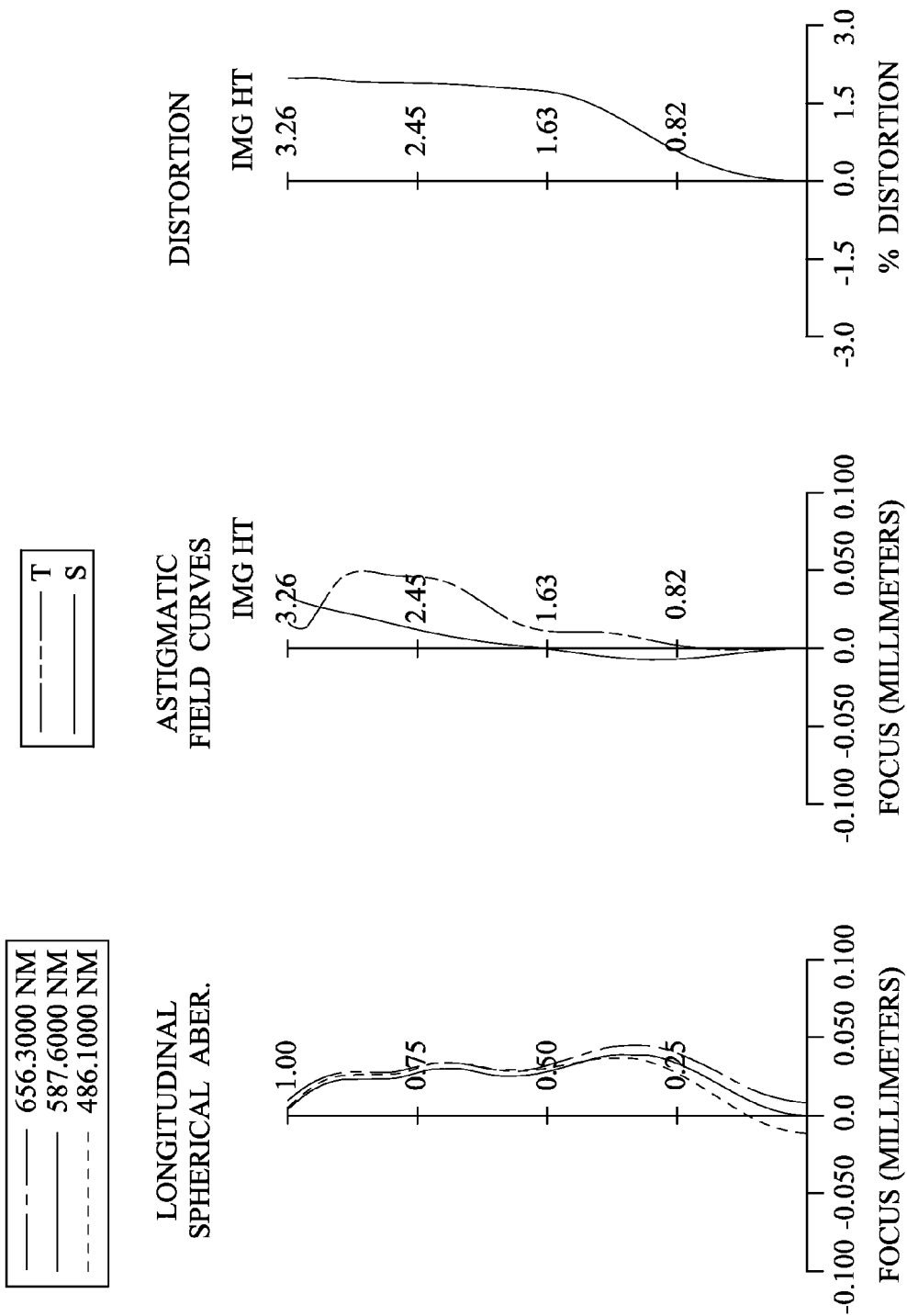
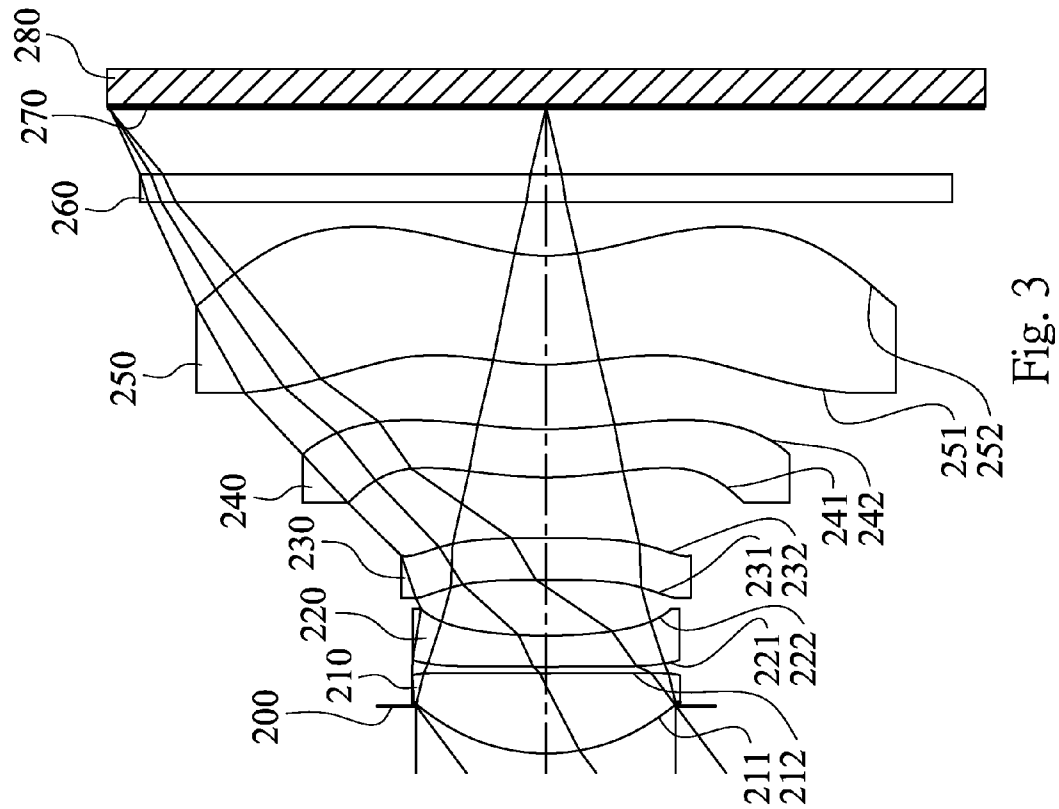


Fig. 2



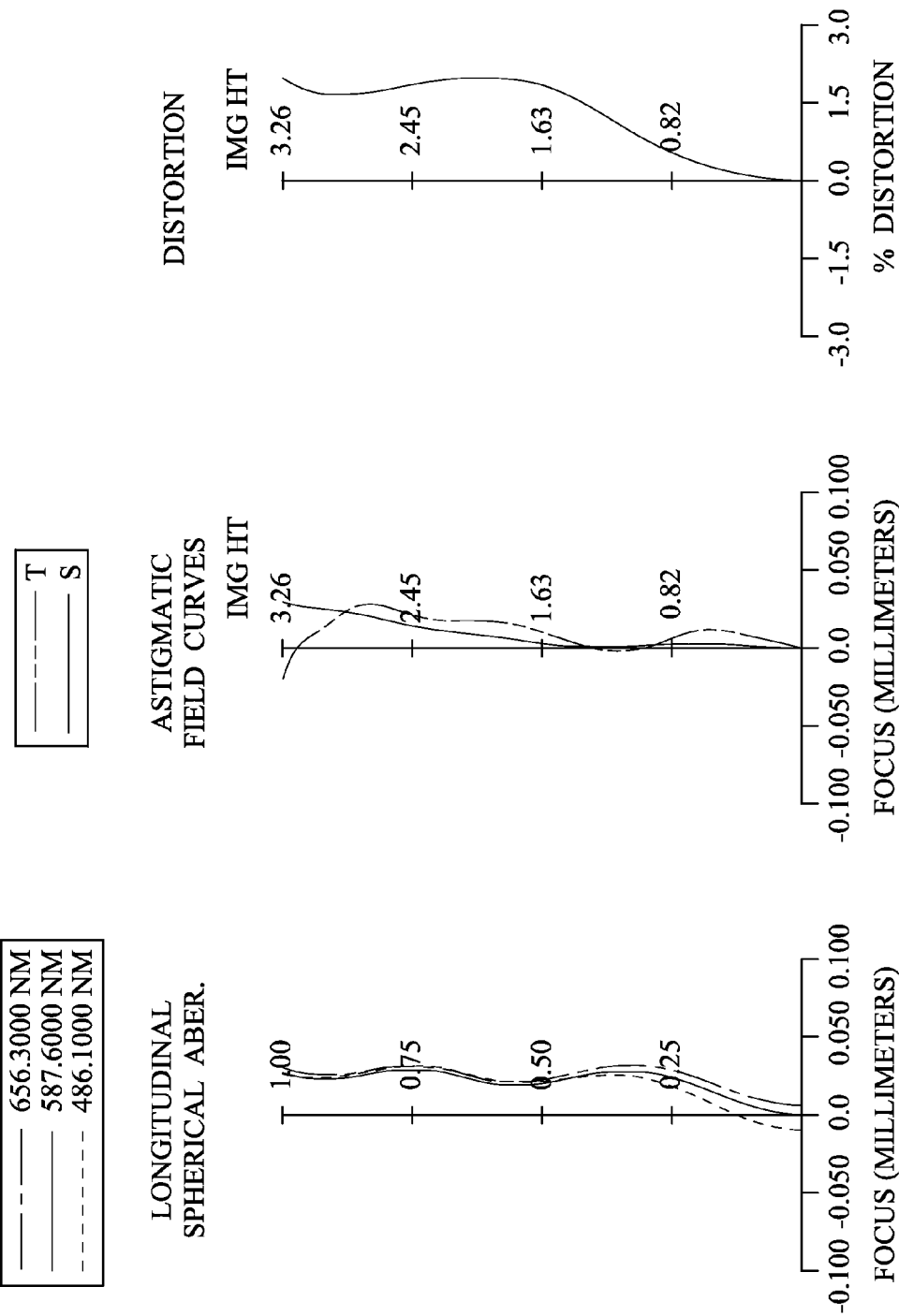


Fig. 4

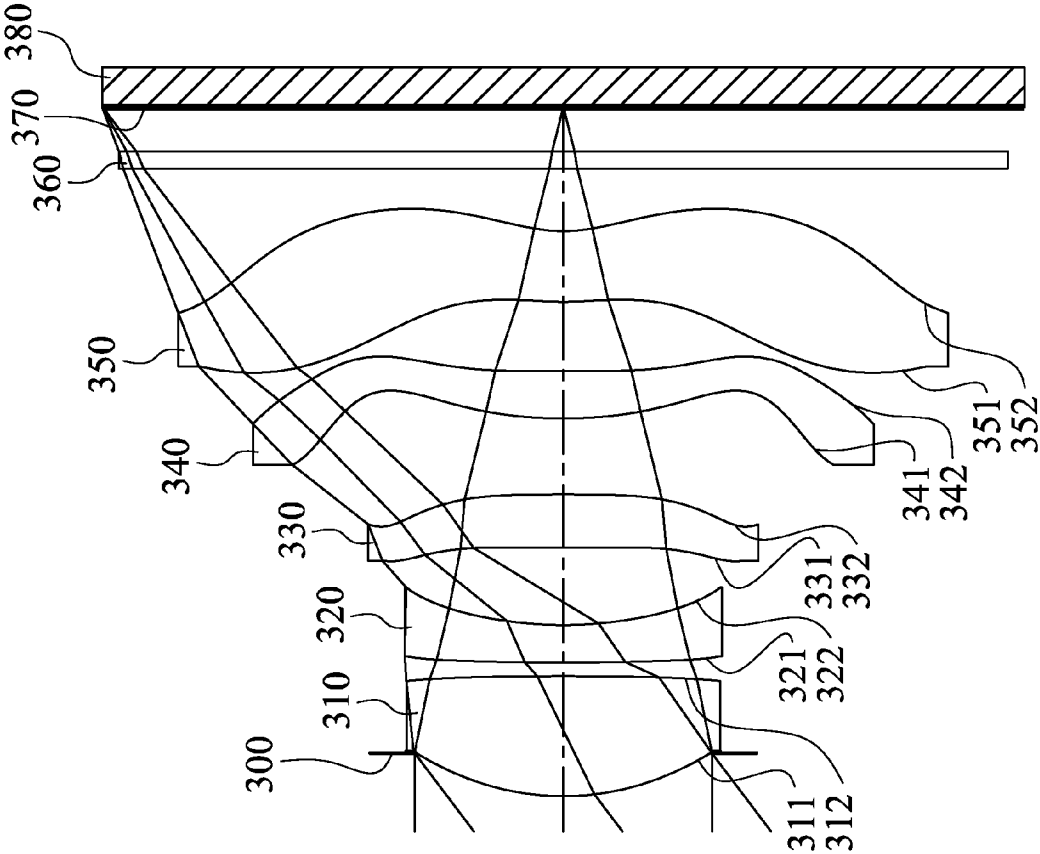
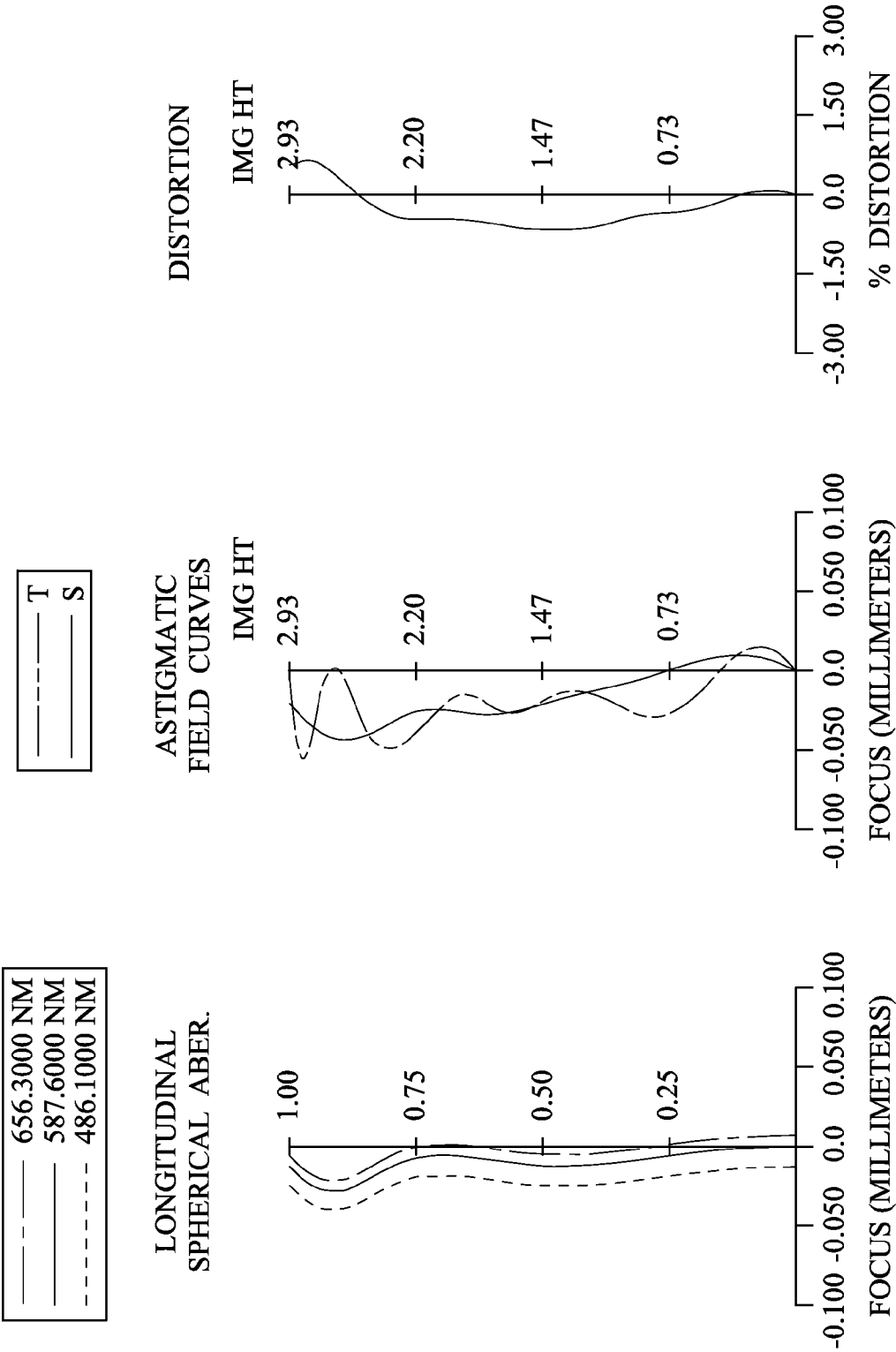


Fig. 5



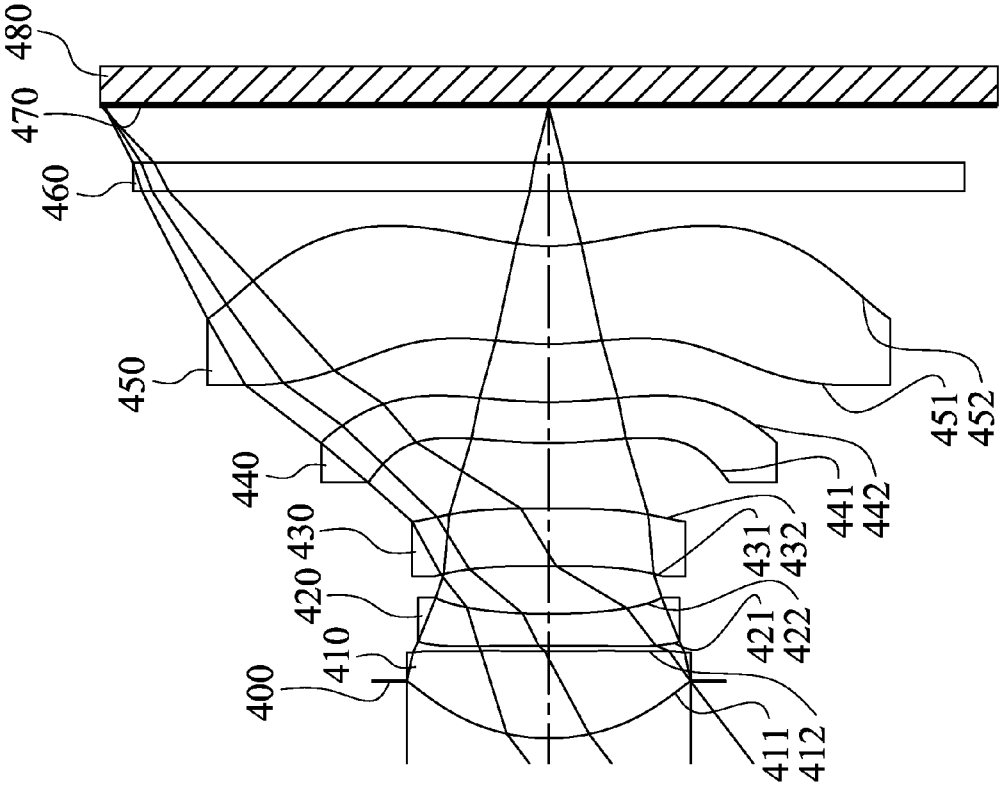


Fig. 7

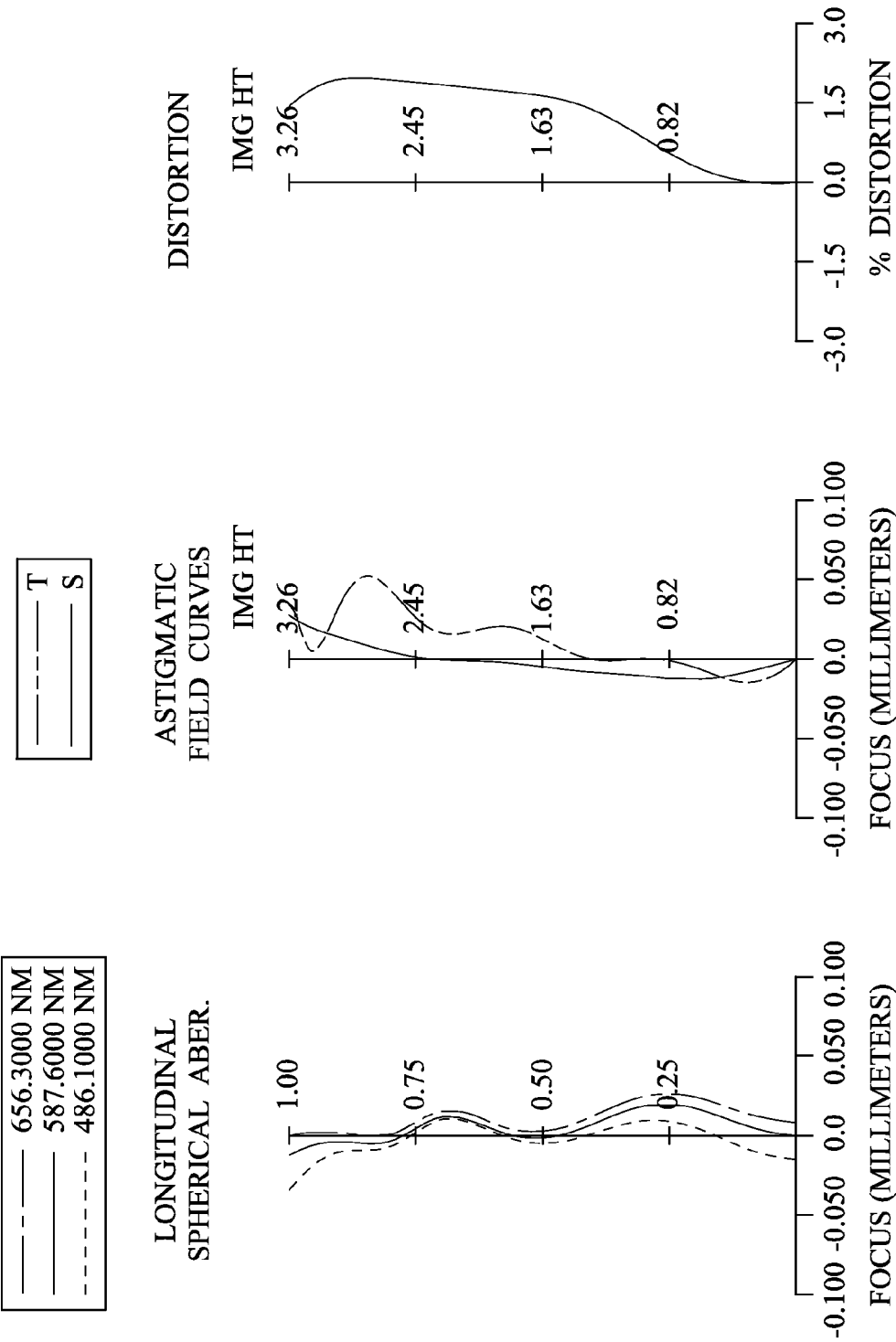


Fig. 8

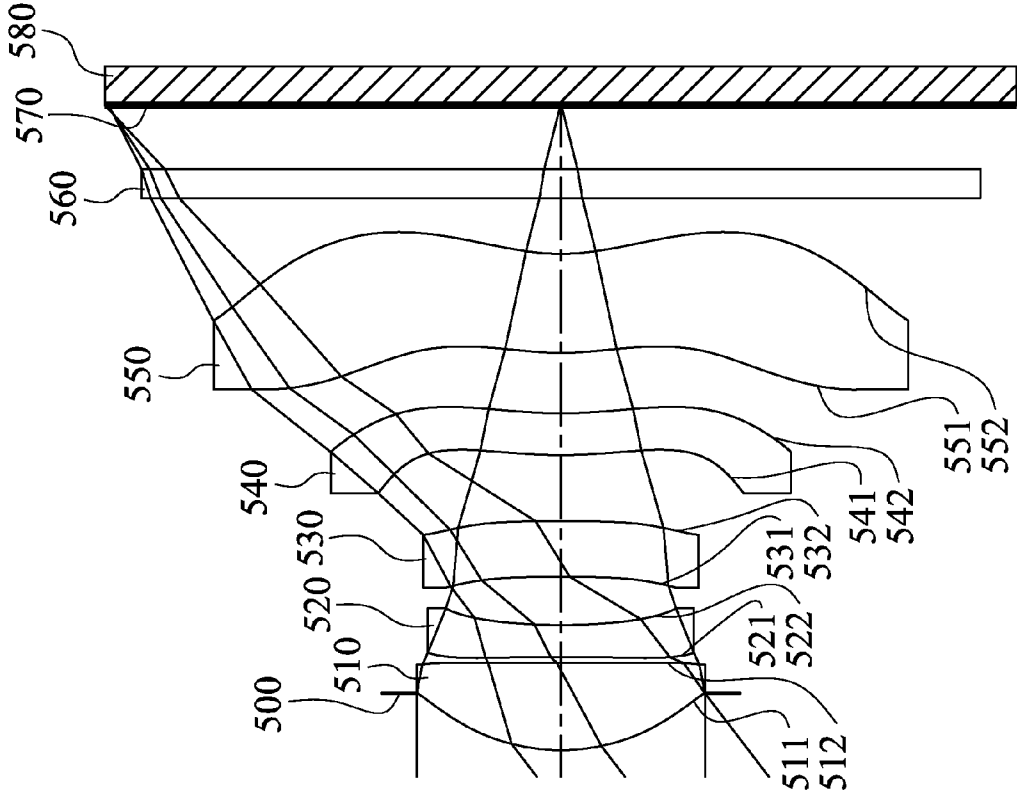
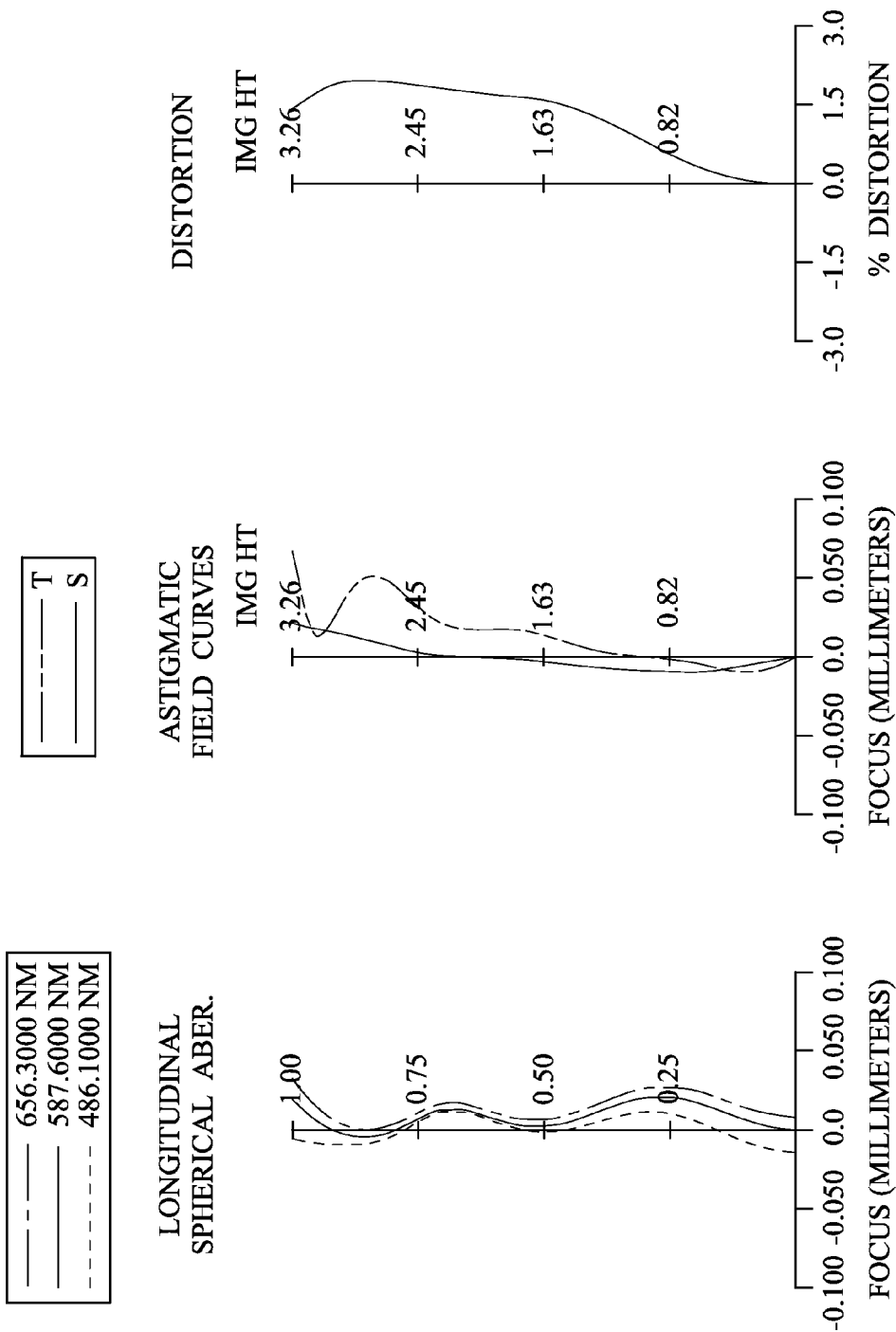


Fig. 9



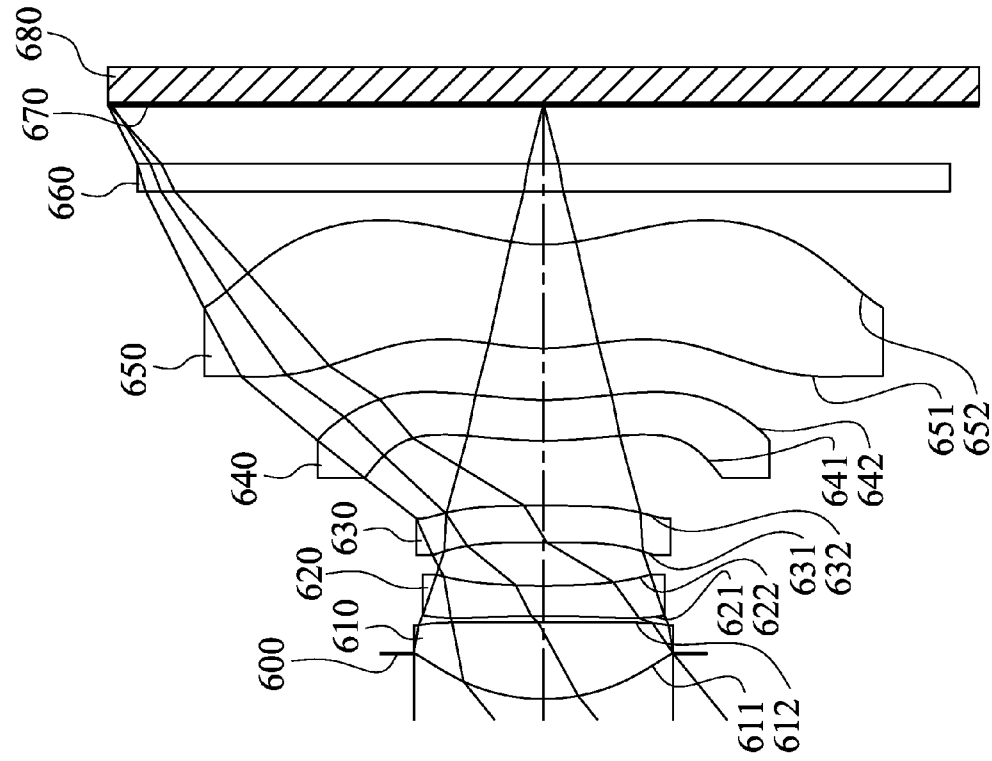
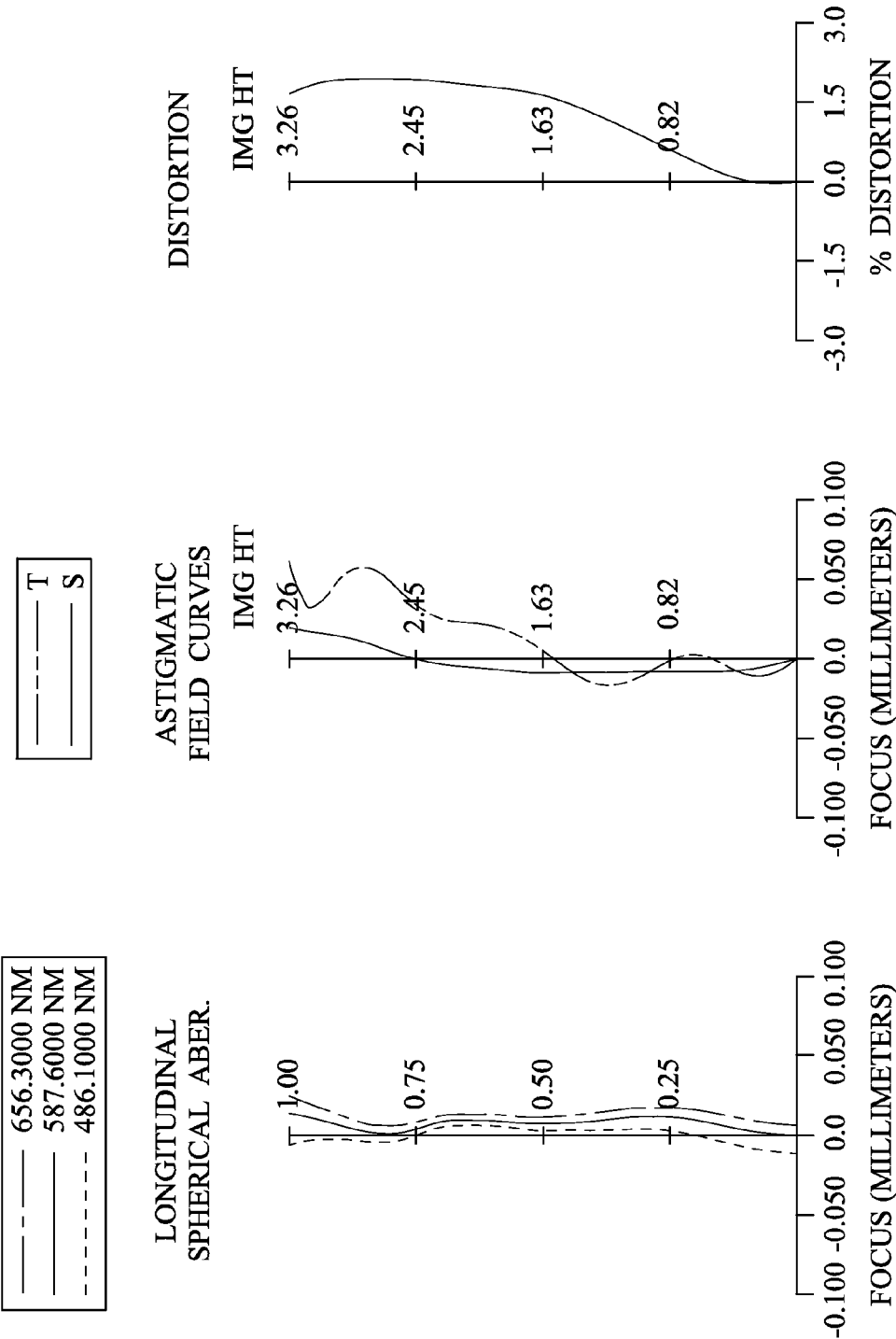


Fig. 11



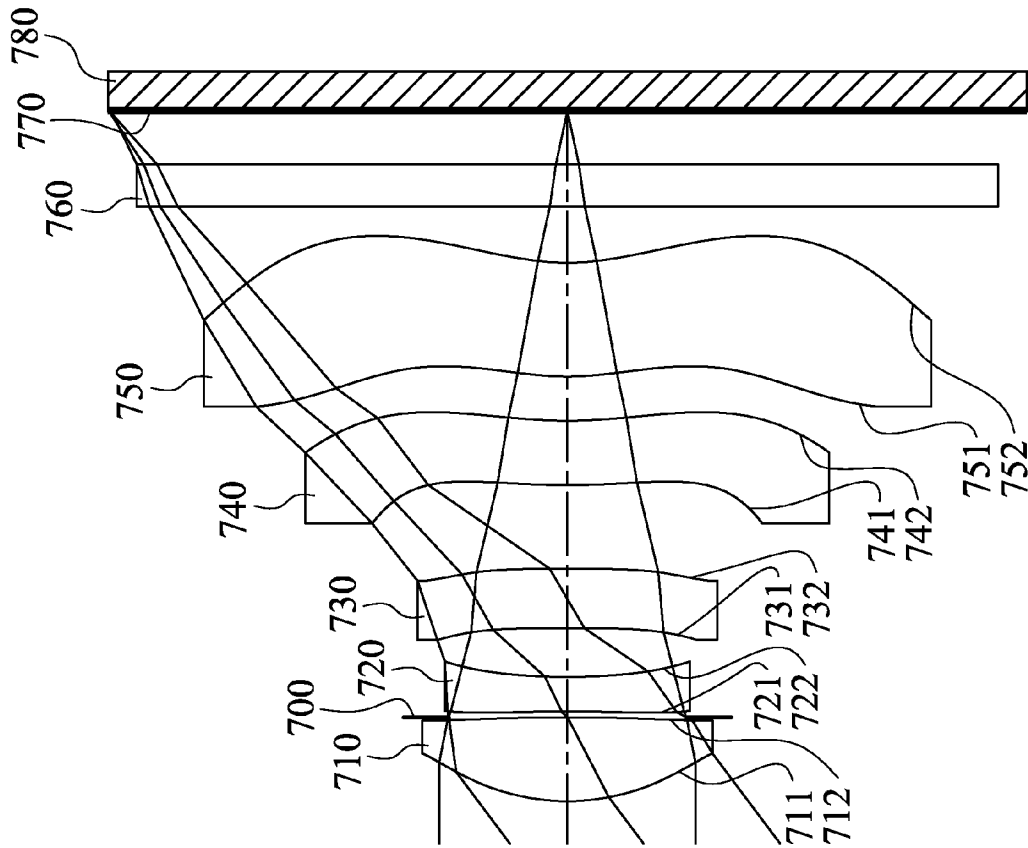


Fig. 13

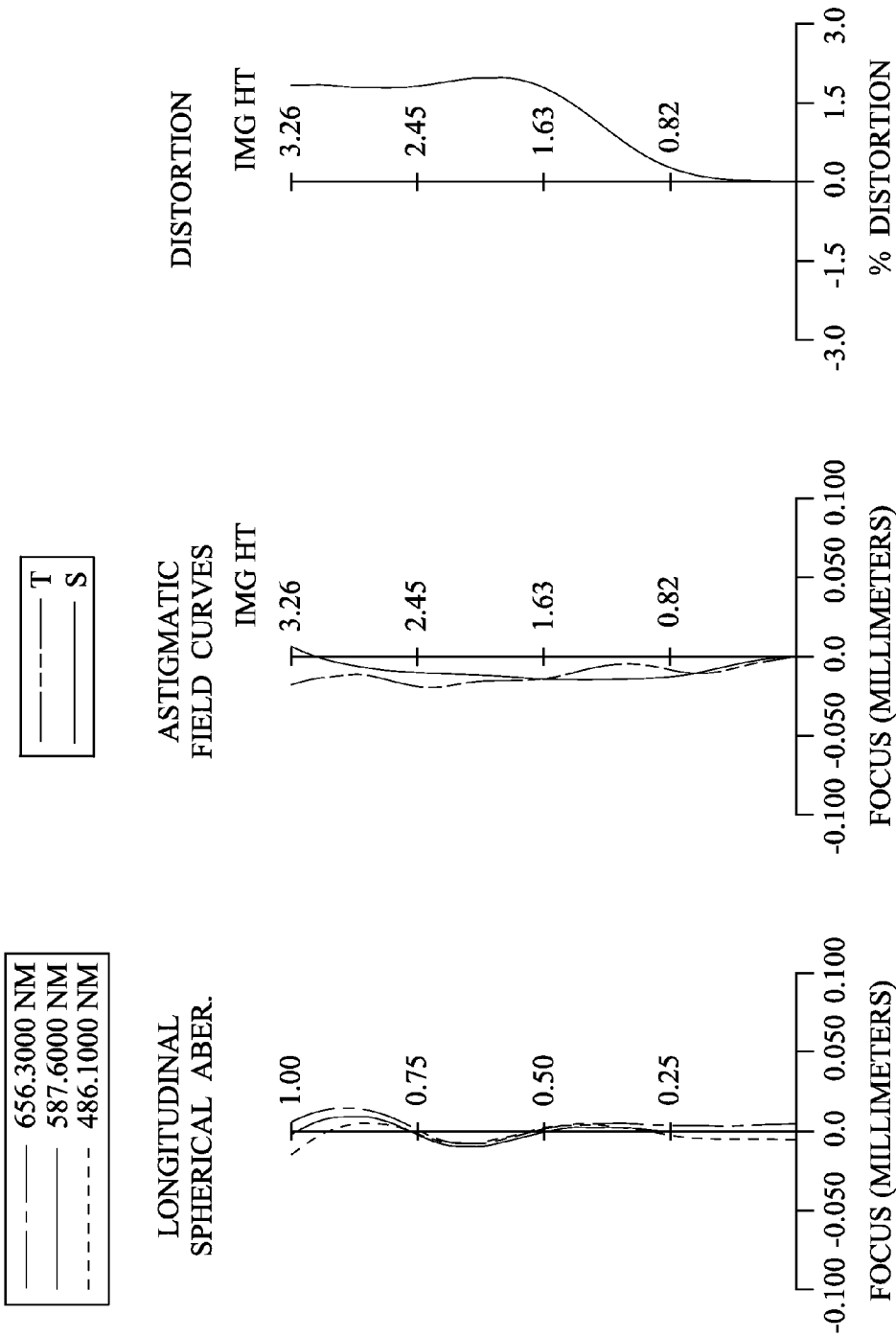


Fig. 14

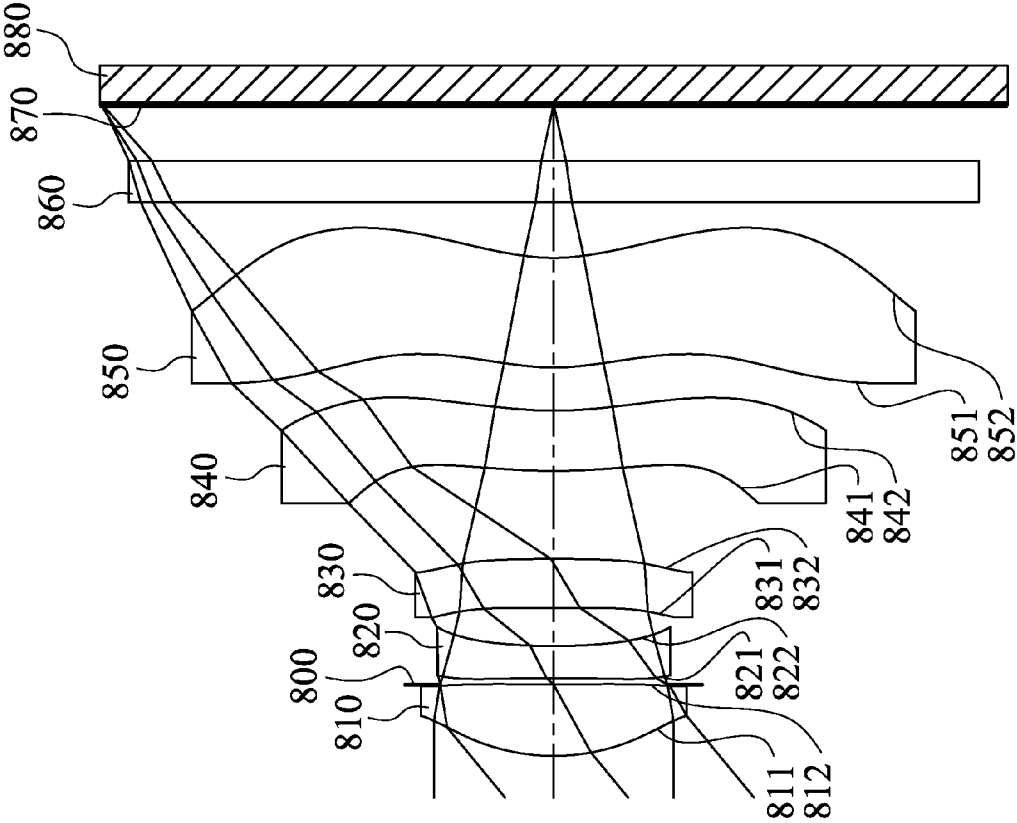


Fig. 15

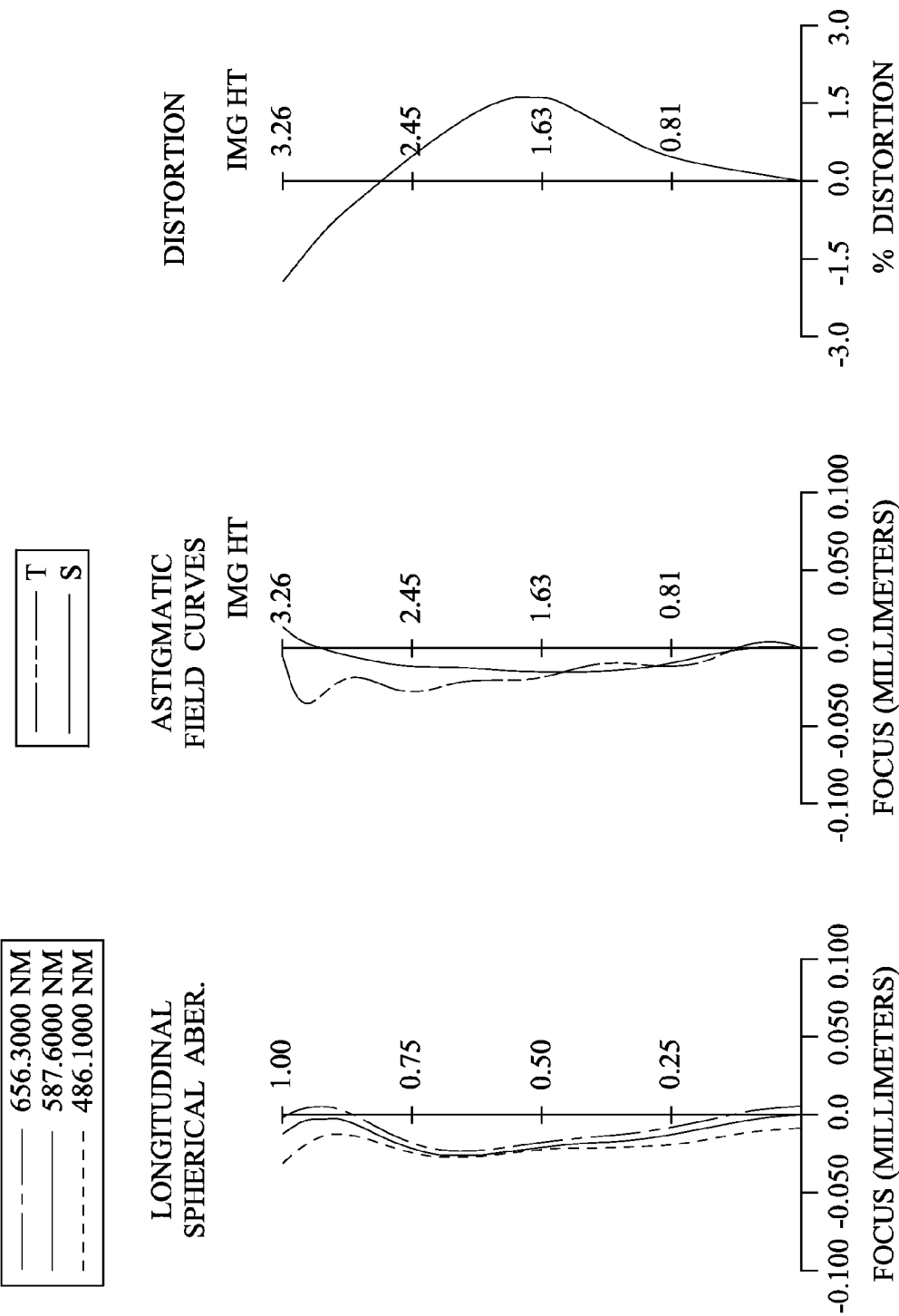


Fig. 16

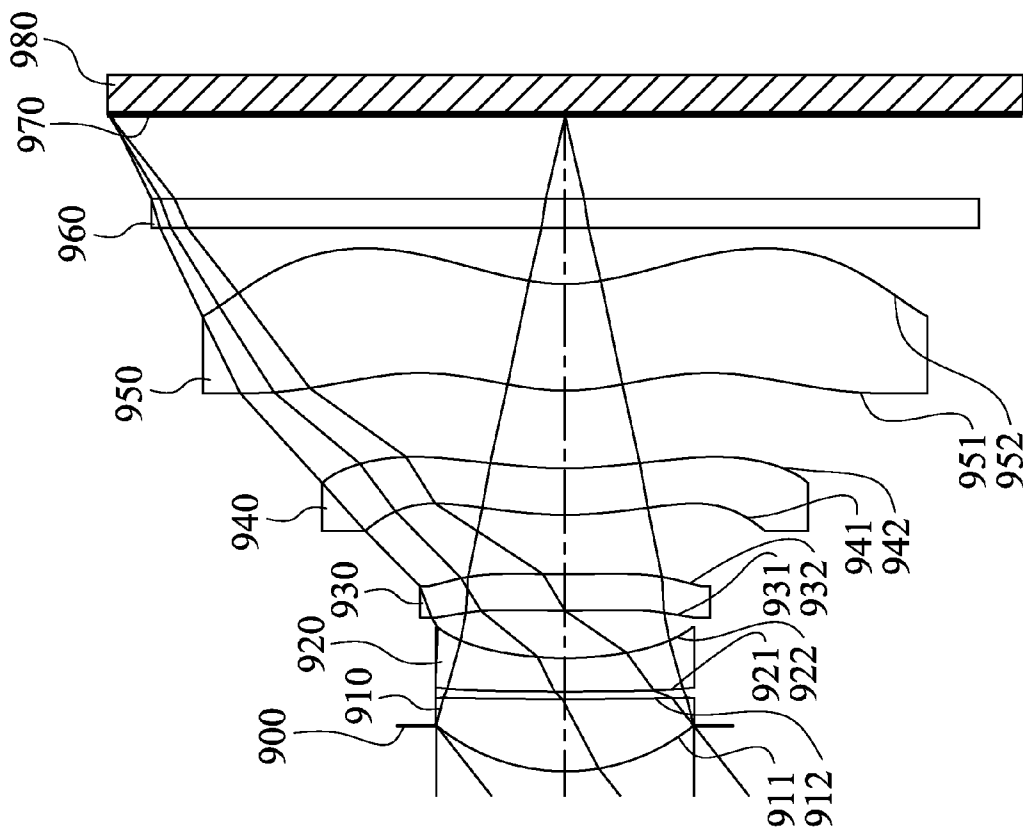
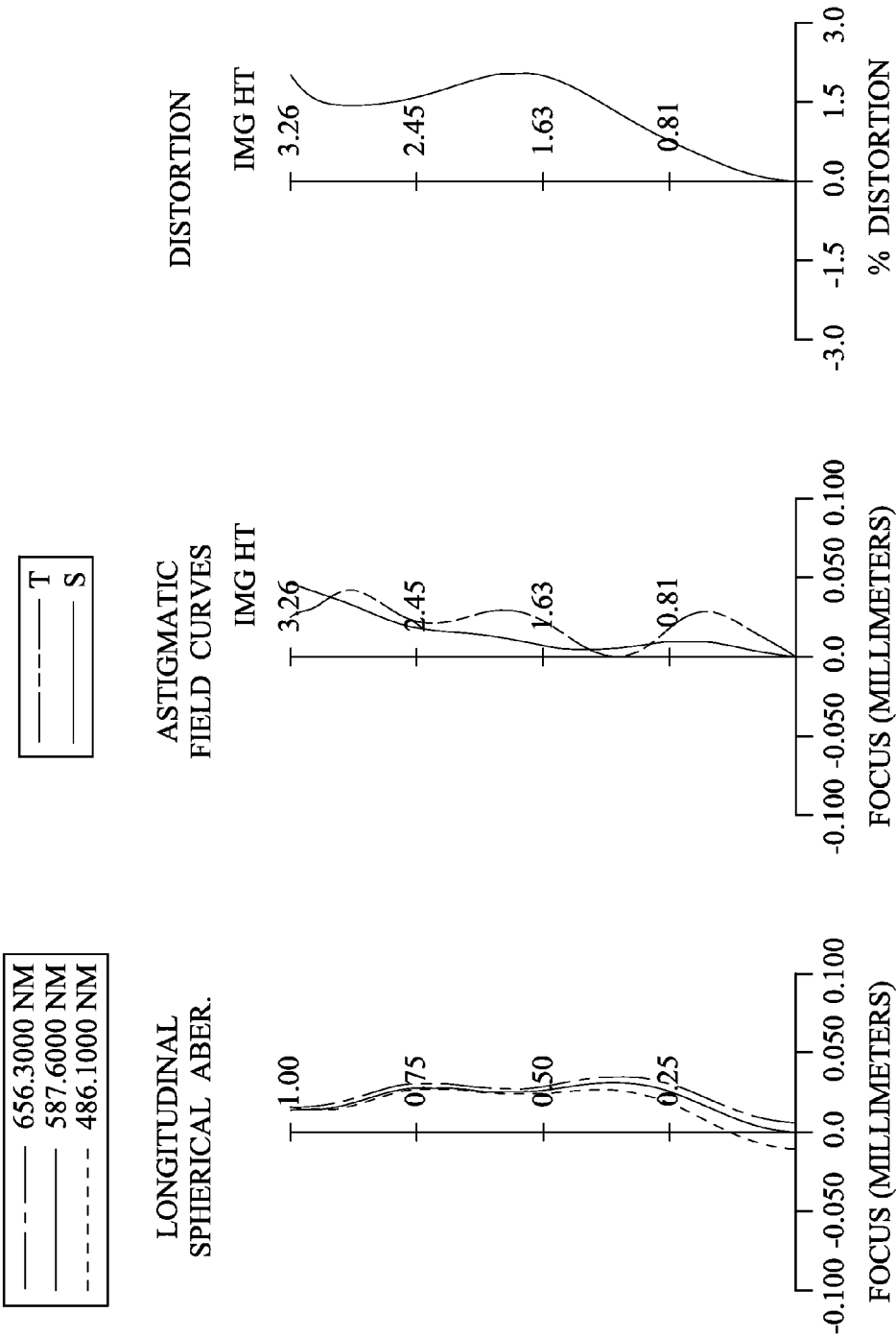


Fig. 17



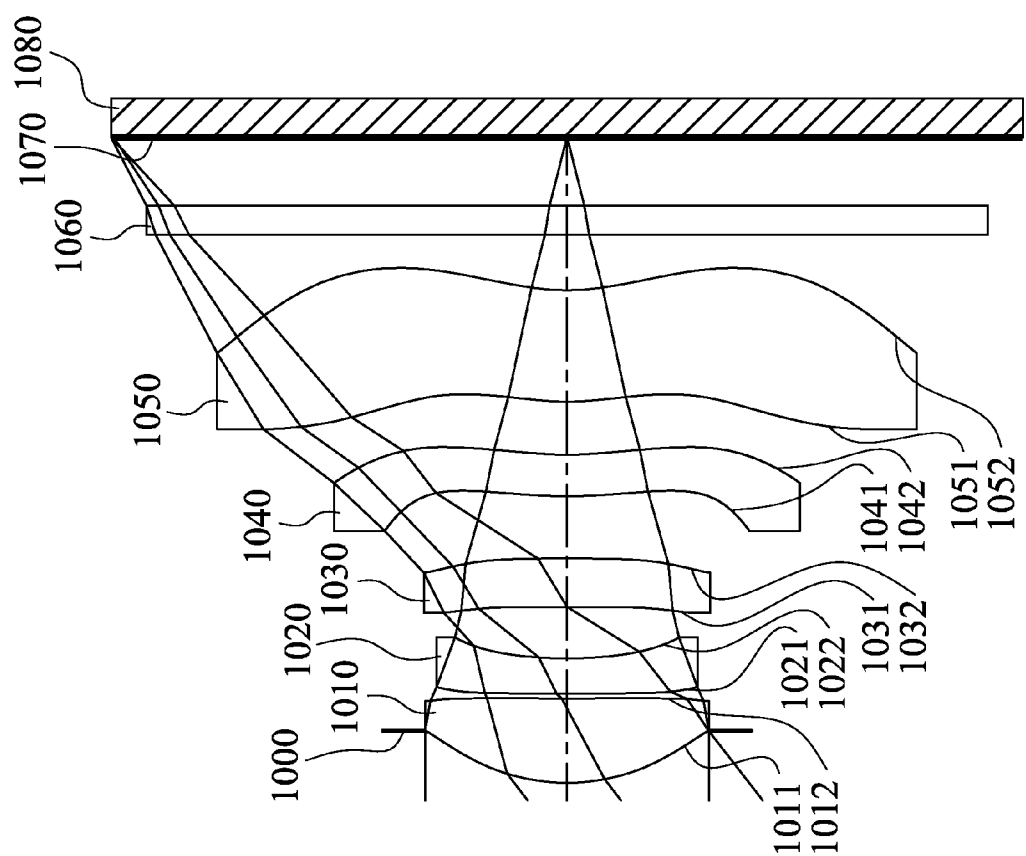


Fig. 19

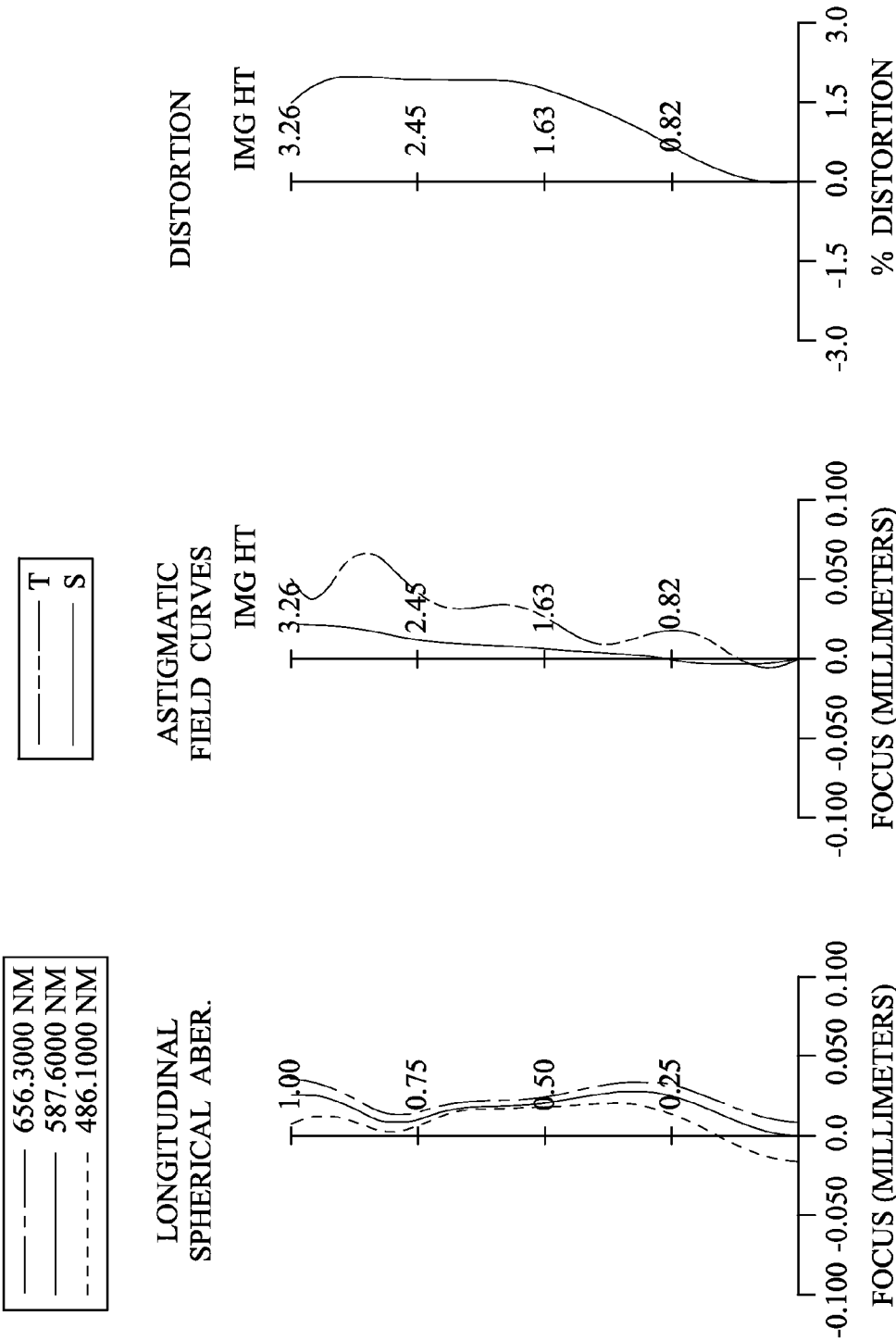


Fig. 20

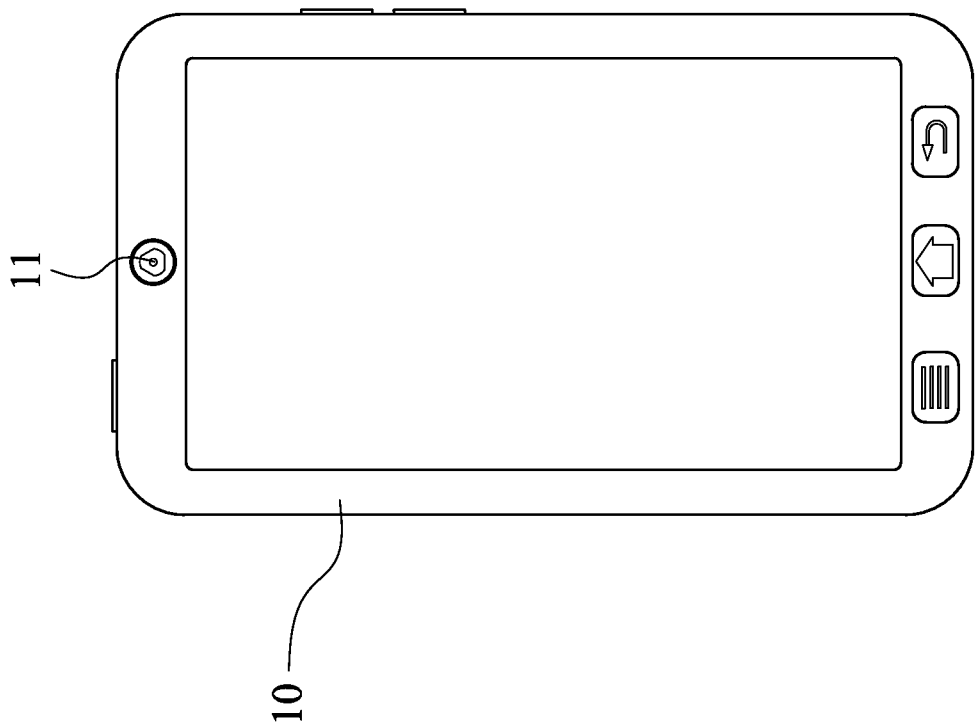


Fig. 21

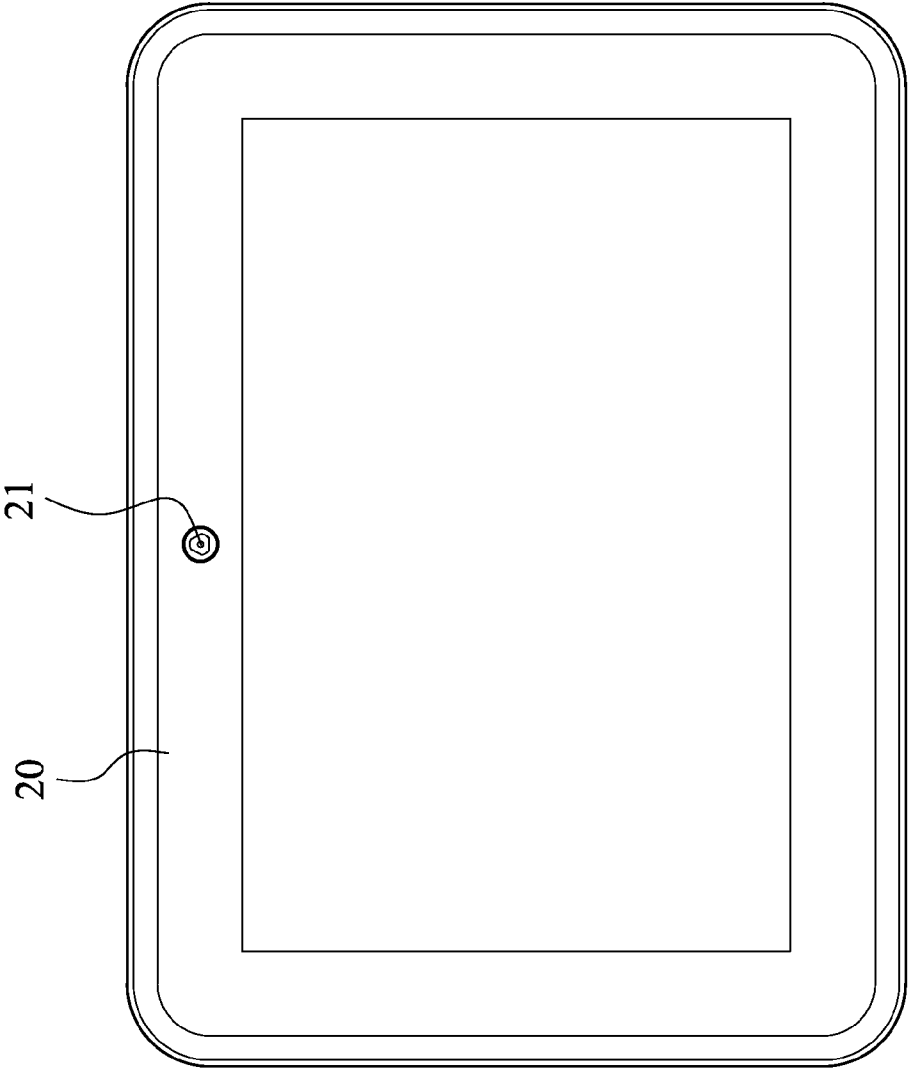


Fig. 22

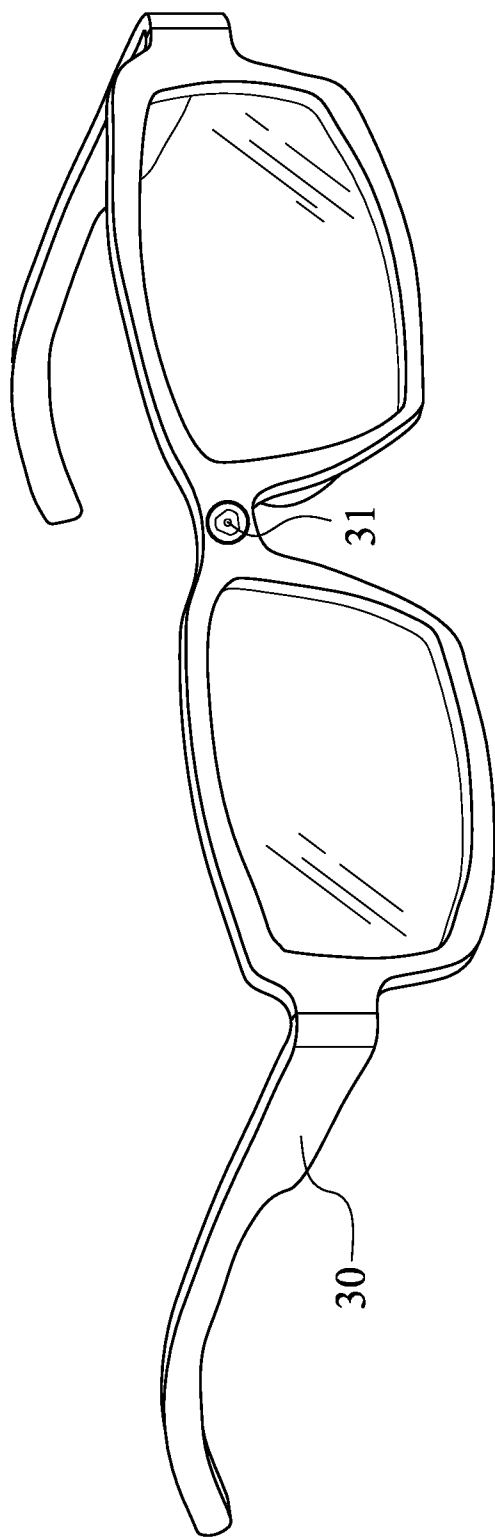


Fig. 23

IMAGING SYSTEM, IMAGE CAPTURING APPARATUS AND ELECTRONIC DEVICE

RELATED APPLICATIONS

[0001] This application claims priority to Taiwan Application Serial Number 104106526, filed Mar. 2, 2015, which is herein incorporated by reference.

BACKGROUND

[0002] 1. Technical Field

[0003] The present disclosure relates to an imaging system and an image capturing apparatus. More particularly, the present disclosure relates to a compact imaging system and an image capturing apparatus which is applicable to electronic devices.

[0004] 2. Description of Related Art

[0005] In recent years, with the popularity of mobile terminals having camera functionalities, the demand of miniaturized optical systems has been increasing. The sensor of a conventional optical system is typically a CCD (Charge-Coupled Device) or a CMOS (Complementary Metal-Oxide-Semiconductor) sensor. As the advanced semiconductor manufacturing technologies have allowed the pixel size of sensors to be reduced and compact optical systems have gradually evolved toward the field of higher megapixels, there is an increasing demand for compact optical systems featuring better image quality.

[0006] Conventional optical systems employed in electronic devices mainly adopt four-element lens structures. The surface shapes of the lens elements of the optical systems with short total track length usually cause low relative illumination in the peripheral regions and poor molding of the lens elements. Hence, the conventional optical systems cannot satisfy these requirements of the compact optical systems.

[0007] Other conventional compact optical systems with five-element lens structure are also developed. However, the excessive thickness variation of the lens elements produces poor molding results, and the high refractive power leads to excessive sensitivity in manufacturing. Furthermore, the large variation of the incident angles also creates problems regarding surface reflections.

SUMMARY

[0008] According to one aspect of the present disclosure, an imaging system includes, in order from an object side to an image side, a first lens element, a second lens element, a third lens element, a fourth lens element and a fifth lens element. The first lens element with positive refractive power has an object-side surface being convex in a paraxial region thereof. The second lens element with negative refractive power has an image-side surface being concave in a paraxial region thereof. The third lens element has refractive power. The fourth lens element with refractive power has an image-side surface being concave in a paraxial region thereof, wherein an object-side surface and the image-side surface of the fourth lens element are aspheric, and the image-side surface of the fourth lens element includes at least one convex shape in an off-axial region thereof. The fifth lens element with refractive power has an object-side surface being convex in a paraxial region thereof and an image-side surface being concave in a paraxial region thereof, wherein the object-side surface and the image-side surface of the fifth lens element are aspheric, and the image-side surface of the fifth lens element includes

at least one convex shape in an off-axial region thereof. The imaging system has a total of five lens elements with refractive power. There is an air gap between any two of the first lens element, the second lens element, the third lens element, the fourth lens element and the fifth lens element that are adjacent to each other. When a central thickness of the fourth lens element is CT4, an axial distance between the third lens element and the fourth lens element is T34, an axial distance between the fourth lens element and the fifth lens element is T45, a focal length of the imaging system is f, a curvature radius of an object-side surface of the third lens element is R5, a curvature radius of an image-side surface of the third lens element is R6, an Abbe number of the second lens element is V2, an Abbe number of the third lens element is V3, and an Abbe number of the fourth lens element is V4, the following conditions are satisfied:

$$CT4/T34 < 1.20;$$

$$CT4/T45 < 1.60;$$

$$|f/R5| + |f/R6| < 1.10; \text{ and}$$

$$1.75 < (V2 + V4)/V3.$$

[0009] According to another aspect of the present disclosure, an image capturing apparatus includes the imaging system according to the aforementioned aspect and an image sensor, wherein the image sensor is disposed on an image surface of the imaging system.

[0010] According to still another aspect of the present disclosure, an electronic device includes the image capturing apparatus according to the foregoing aspect.

BRIEF DESCRIPTION OF THE DRAWINGS

[0011] The disclosure can be more fully understood by reading the following detailed description of the embodiments, with reference made to the accompanying drawings as follows:

[0012] FIG. 1 is a schematic view of an image capturing apparatus according to the 1st embodiment of the present disclosure;

[0013] FIG. 2 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing apparatus according to the 1st embodiment;

[0014] FIG. 3 is a schematic view of an image capturing apparatus according to the 2nd embodiment of the present disclosure;

[0015] FIG. 4 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing apparatus according to the 2nd embodiment;

[0016] FIG. 5 is a schematic view of an image capturing apparatus according to the 3rd embodiment of the present disclosure;

[0017] FIG. 6 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing apparatus according to the 3rd embodiment;

[0018] FIG. 7 is a schematic view of an image capturing apparatus according to the 4th embodiment of the present disclosure;

[0019] FIG. 8 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing apparatus according to the 4th embodiment;

[0020] FIG. 9 is a schematic view of an image capturing apparatus according to the 5th embodiment of the present disclosure;

[0021] FIG. 10 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing apparatus according to the 5th embodiment;

[0022] FIG. 11 is a schematic view of an image capturing apparatus according to the 6th embodiment of the present disclosure;

[0023] FIG. 12 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing apparatus according to the 6th embodiment;

[0024] FIG. 13 is a schematic view of an image capturing apparatus according to the 7th embodiment of the present disclosure;

[0025] FIG. 14 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing apparatus according to the 7th embodiment;

[0026] FIG. 15 is a schematic view of an image capturing apparatus according to the 8th embodiment of the present disclosure;

[0027] FIG. 16 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing apparatus according to the 8th embodiment;

[0028] FIG. 17 is a schematic view of an image capturing apparatus according to the 9th embodiment of the present disclosure;

[0029] FIG. 18 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing apparatus according to the 9th embodiment;

[0030] FIG. 19 is a schematic view of an image capturing apparatus according to the 10th embodiment of the present disclosure;

[0031] FIG. 20 shows spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing apparatus according to the 10th embodiment;

[0032] FIG. 21 shows an electronic device according to the 11th embodiment of the present disclosure;

[0033] FIG. 22 shows an electronic device according to the 12th embodiment of the present disclosure; and

[0034] FIG. 23 shows an electronic device according to the 13th embodiment of the present disclosure.

DETAILED DESCRIPTION

[0035] An imaging system includes, in order from an object side to an image side, a first lens element, a second lens element, a third lens element, a fourth lens element and a fifth lens element. The imaging system has a total of five lens elements with refractive power.

[0036] According to the imaging system of the present disclosure, there is an air gap between any two of the first lens element, the second lens element, the third lens element, the fourth lens element and the fifth lens element that are adjacent to each other, that is, each of the first through fifth lens elements of the imaging system is a single and non-cemented lens element. Moreover, the manufacturing process of the cemented lenses is more complex than the non-cemented lenses. In particular, an image-side surface of one lens element and an object-side surface of the following lens element need to have accurate curvature to ensure these two lens elements will be highly cemented. However, during the cementing process, those two lens elements might not be highly cemented due to displacement and it is thereby not favorable for the image quality of the imaging system. Therefore, there is an air gap between any two of the first lens element, the second lens element, the third lens element, the fourth lens element and the fifth lens element that are adjacent

to each other in the present disclosure for resolving the problem generated by the cemented lens elements.

[0037] The first lens element with positive refractive power has an object-side surface being convex in a paraxial region thereof, and can have an image-side surface being concave in a paraxial region thereof. Therefore, it is favorable for reducing the total track length of the imaging system by properly adjusting the positive refractive power of the first lens element.

[0038] The second lens element with negative refractive power has an image-side surface being concave in a paraxial region thereof. Therefore, it is favorable for effectively correcting the aberration of the imaging system.

[0039] The third lens element can have positive refractive power. Therefore, it is favorable for reducing the sensitivity of refractive power distribution in the imaging system.

[0040] The fourth lens element can have an object-side surface being convex in a paraxial region thereof, and has an image-side surface being concave in a paraxial region thereof, wherein the image-side surface of the fourth lens element includes at least one convex shape in an off-axial region thereof. Therefore, it is favorable for balancing the thickness arrangement between the paraxial region and the off-axial region of the fourth lens element so as to improve upon the poor molding results from excessive lens thickness variation.

[0041] The fifth lens element has an object-side surface being convex in a paraxial region thereof and an image-side surface being concave in a paraxial region thereof, wherein the image-side surface of the fifth lens element includes at least one convex shape in an off-axial region thereof. Therefore, it is favorable for balancing the thickness arrangement between the paraxial region and the off-axial region of the fifth lens element so as to improve upon the poor molding results from excessive lens thickness variation.

[0042] When a central thickness of the fourth lens element is CT_4 , and an axial distance between the third lens element and the fourth lens element is T_{34} , the following condition is satisfied: $CT_4/T_{34} < 1.20$. Therefore, it is favorable for arranging the third lens element and the fourth lens element so as to reduce the collision probability among the lens elements during assembling. Preferably, the following condition is satisfied: $CT_4/T_{34} < 1.0$.

[0043] When the central thickness of the fourth lens element is CT_4 , and an axial distance between the fourth lens element and the fifth lens element is T_{45} , the following condition is satisfied: $CT_4/T_{45} < 1.60$. Therefore, it is favorable for arranging the fourth lens element and the fifth lens element so as to reduce the collision probability among the lens elements during assembling.

[0044] Preferably, the following condition is satisfied: $CT_4/T_{45} < 1.48$. More preferably, the following condition is satisfied: $CT_4/T_{45} < 1.25$.

[0045] When a focal length of the imaging system is f , a curvature radius of an object-side surface of the third lens element is R_5 , and a curvature radius of an image-side surface of the third lens element is R_6 , the following condition is satisfied: $|f/R_5| + |f/R_6| < 1.10$. Therefore, it is favorable for ensuring the third lens element with lesser refractive power in a paraxial region thereof so as to solve the problems of the image center excessively corrected. Preferably, the following condition is satisfied: $|f/R_5| + |f/R_6| < 0.75$.

[0046] When an Abbe number of the second lens element is V_2 , an Abbe number of the third lens element is V_3 , and an Abbe number of the fourth lens element is V_4 , the following

condition is satisfied: $1.75 < (V2+V4)/V3$. Therefore, it is favorable for balancing the correction in chromatic aberration and the high-order aberration.

[0047] When the Abbe number of the third lens element is $V3$, the following condition is satisfied: $V3 < 30$. Therefore, it is favorable for correcting the chromatic aberration of the imaging system.

[0048] When a curvature radius of the object-side surface of the fourth lens element is $R7$, and the focal length of the imaging system is f , the following condition is satisfied: $|R7/f| < 3.0$. Therefore, it is favorable for correcting the aberration of the imaging system and improving the image quality. Preferably, the following condition is satisfied: $|R7/f| < 2.3$.

[0049] When an f-number of the imaging system is Fno , the following condition is satisfied: $1.6 < Fno < 2.4$. Therefore, it is favorable for obtaining a large aperture so as to capture clear image under insufficient light conditions.

[0050] When an axial distance between the object-side surface of the first lens element and an image surface is TL , the following condition is satisfied: $TL < 6$ mm. Therefore, it is favorable for reducing the total track length of the imaging system so as to maintain the compact size thereof.

[0051] When half of a maximal field of view of the imaging system is $HFOV$, the following condition is satisfied: 35 degrees $< HFOV$. Therefore, it is favorable for obtaining a proper field of view and a greater imaging scene.

[0052] When the focal length of the imaging system is f , a focal length of the third lens element is $f3$, a focal length of the fourth lens element is $f4$, and a focal length of the fifth lens element is $f5$, the following condition is satisfied: $|f/f3| + |f/f4| + |f/f5| < 0.60$. Therefore, it is favorable for reducing the sensitivity of refractive power distribution so as to prevent the problems of surface reflections from the excessive change of the incident angles.

[0053] When a focal length of the first lens element is $f1$, a focal length of the second lens element is $f2$, the focal length of the third lens element is $f3$, the focal length of the fourth lens element is $f4$, and the focal length of the fifth lens element is $f5$, the following conditions are satisfied: $|f3| > |f1|$; $|f3| > |f2|$; $|f3| > |f4|$; and $|f3| > |f5|$. Therefore, it is favorable for balancing the refractive power distribution of the imaging system so as to reduce the sensitivity of refractive power distribution.

[0054] When the curvature radius of the object-side surface of the fourth lens element is $R7$, a curvature radius of the image-side surface of the fourth lens element is $R8$, a curvature radius of the object-side surface of the fifth lens element is $R9$, a curvature radius of the image-side surface of the fifth lens element is $R10$, and the focal length of the imaging system is f , the following condition is satisfied: $(|R7| + |R8| + |R9| + |R10|)/f < 3.5$. Therefore, it is favorable for properly arranging the surface shapes of the fourth lens element and the fifth lens element so as to solve the problems of poor molding.

[0055] When an axial distance between the first lens element and the second lens element is $T12$, an axial distance between the second lens element and the third lens element is $T23$, the axial distance between the third lens element and the fourth lens element is $T34$, the axial distance between the fourth lens element and the fifth lens element is $T45$, and a sum of axial distances between any two of the first lens element, the second lens element, the third lens element, the fourth lens element and the fifth lens element that are adjacent to each other is ΣAT ($\Sigma AT = T12 + T23 + T34 + T45$), the follow-

ing condition is satisfied: $0.60 < (T34 + T45)/\Sigma AT < 0.80$. Therefore, it is favorable for assembling the lens elements so as to increase the manufacturing yield rate.

[0056] When the Abbe number of the second lens element is $V2$, the Abbe number of the third lens element is $V3$, and the Abbe number of the fourth lens element is $V4$, the following condition is satisfied: $V2 + V3 + V4 < 90$. Therefore, it is favorable for correcting the chromatic aberration of the imaging system.

[0057] When a central thickness of the first lens element is $CT1$, a central thickness of the second lens element is $CT2$, a central thickness of the third lens element is $CT3$, the central thickness of the fourth lens element is $CT4$, and a central thickness of the fifth lens element is $CT5$, $CT5$ is larger than $CT1$, $CT2$, $CT3$, and $CT4$. Therefore, it is favorable for the moldability and the homogeneity of lens elements so as to increase the manufacturing yield rate.

[0058] When the central thickness of the second lens element is $CT2$, the central thickness of the third lens element is $CT3$, the central thickness of the fourth lens element is $CT4$, and the central thickness of the fifth lens element is $CT5$, the following condition is satisfied: $(CT2 + CT3 + CT4)/CT5 < 1.25$. Therefore, it is favorable for the moldability and the homogeneity of lens elements.

[0059] When the curvature radius of the image-side surface of the fourth lens element is $R8$, and the curvature radius of the object-side surface of the fifth lens element is $R9$, the following condition is satisfied: $1.20 < R8/R9$. Therefore, it is favorable for effectively correcting the aberration.

[0060] According to the imaging system of the present disclosure, the lens elements thereof can be made of plastic or glass material. When the lens elements are made of plastic material, the manufacturing costs can be effectively reduced. When the lens elements are made of glass material, the arrangement of the refractive power of the imaging system may be more flexible to design. Furthermore, surfaces of each lens element can be arranged to be aspheric, since the aspheric surface of the lens element is easy to form a shape other than spherical surface so as to have more controllable variables for eliminating the aberration thereof, and to further decrease the required number of the lens elements. Therefore, the total track length of the imaging system can also be reduced.

[0061] According to the imaging system of the present disclosure, each of an object-side surface and an image-side surface has a paraxial region and an off-axis region. The paraxial region refers to the region of the surface where light rays travel close to the optical axis, and the off-axis region refers to the region of the surface away from the paraxial region. Particularly, when the lens element has a convex surface, it indicates that the surface is convex in the paraxial region thereof; when the lens element has a concave surface, it indicates that the surface is concave in the paraxial region thereof. Furthermore, when the lens element has positive refractive power or negative refractive power, it indicates that the lens element has refractive power in the paraxial region thereof. When the lens element has a focal length, it indicates that the lens element has a focal length in the paraxial region thereof.

[0062] According to the imaging system of the present disclosure, the image surface, depending on the corresponding image sensor, can be a planar surface or a curved surface

with any curvature. When the image surface is a curved surface, it is particularly indicates a concave surface toward the object side.

[0063] According to the imaging system of the present disclosure, an aperture stop can be configured as a front stop or a middle stop. A front stop disposed between an imaged object and the first lens element can provide a longer distance between an exit pupil of the imaging system and the image surface and thereby improves the image-sensing efficiency of an image sensor. A middle stop disposed between the first lens element and the image surface is favorable for enlarging the field of view of the imaging system and thereby provides a wider field of view for the same.

[0064] According to the imaging system of the present disclosure, the imaging system can include at least one stop, such as an aperture stop, a glare stop or a field stop. The glare stop or the field stop is for eliminating the stray light and thereby improving the image resolution thereof.

[0065] According to the imaging system of the present disclosure, the imaging system can be optionally applied to moving focus optical systems. Furthermore, the imaging system is featured with good correction ability and high image quality, and can be applied to 3D (three-dimensional) image capturing applications, in products such as digital cameras, mobile devices, digital tablets, smart TVs, network monitoring devices, motion sensing input devices, driving recorders, rear view camera systems, wearable devices and other electronic imaging products.

[0066] According to the present disclosure, an image capturing apparatus is provided. The image capturing apparatus includes the aforementioned imaging system according to the present disclosure and an image sensor, wherein the image sensor is disposed on or near an image surface of the aforementioned imaging system. In the imaging system of the image capturing apparatus, it is favorable for balancing the thickness between the paraxial regions and the off-axial regions of the fourth lens element and the fifth lens element so as to reduce molding problems from excessive lens thickness variation. Moreover, it is favorable for arranging the third lens element, the fourth lens element and the fifth lens element so as to reduce the collision probability among the lens elements during assembling. In addition, it is favorable for solving the problems of the image center excessively corrected and balancing the correction of chromatic aberration and the high-order aberration. Preferably, the image capturing apparatus can further include a barrel member, a holder member or a combination thereof.

[0067] According to the present disclosure, an electronic device is provided, wherein the electronic device includes the aforementioned image capturing apparatus. Therefore, it is favorable for molding, assembling, obtaining proper refractive power and satisfying the requirements of relative illumination in the peripheral regions of the imaging system with short total track length. Preferably, the electronic device can further include but not limited to a control unit, a display, a storage unit, a random access memory unit (RAM) or a combination thereof.

[0068] According to the above description of the present disclosure, the following 1st-13th specific embodiments are provided for further explanation.

1st Embodiment

[0069] FIG. 1 is a schematic view of an image capturing apparatus according to the 1st embodiment of the present

disclosure. FIG. 2 shows, in order from left to right, spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing apparatus according to the 1st embodiment.

[0070] In FIG. 1, the image capturing apparatus includes the imaging system (its reference numeral is omitted) and an image sensor 180. The imaging system includes, in order from an object side to an image side, an aperture stop 100, a first lens element 110, a second lens element 120, a third lens element 130, a fourth lens element 140, a fifth lens element 150, an IR-cut filter 160 and an image surface 170. The image sensor 180 is disposed on the image surface 170 of the imaging system. The imaging system has a total of five lens elements (110-150) with refractive power. Moreover, there is an air gap between any two of the first lens element 110, the second lens element 120, the third lens element 130, the fourth lens element 140 and the fifth lens element 150 that are adjacent to each other.

[0071] The first lens element 110 with positive refractive power has an object-side surface 111 being convex in a paraxial region thereof and an image-side surface 112 being concave in a paraxial region thereof. The first lens element 110 is made of plastic material and has the object-side surface 111 and the image-side surface 112 being both aspheric.

[0072] The second lens element 120 with negative refractive power has an object-side surface 121 being convex in a paraxial region thereof and an image-side surface 122 being concave in a paraxial region thereof. The second lens element 120 is made of plastic material and has the object-side surface 121 and the image-side surface 122 being both aspheric.

[0073] The third lens element 130 with negative refractive power has an object-side surface 131 being concave in a paraxial region thereof and an image-side surface 132 being convex in a paraxial region thereof. The third lens element 130 is made of plastic material and has the object-side surface 131 and the image-side surface 132 being both aspheric.

[0074] The fourth lens element 140 with positive refractive power has an object-side surface 141 being convex in a paraxial region thereof and an image-side surface 142 being concave in a paraxial region thereof. The fourth lens element 140 is made of plastic material and has the object-side surface 141 and the image-side surface 142 being both aspheric. Furthermore, the image-side surface 142 of the fourth lens element 140 includes at least one convex shape in an off-axial region thereof.

[0075] The fifth lens element 150 with negative refractive power has an object-side surface 151 being convex in a paraxial region thereof and an image-side surface 152 being concave in a paraxial region thereof. The fifth lens element 150 is made of plastic material and has the object-side surface 151 and the image-side surface 152 being both aspheric. Furthermore, the image-side surface 152 of the fifth lens element 150 includes at least one convex shape in an off-axial region thereof.

[0076] The IR-cut filter 160 is made of glass material and located between the fifth lens element 150 and the image surface 170, and will not affect the focal length of the imaging system.

[0077] The equation of the aspheric surface profiles of the aforementioned lens elements of the 1st embodiment is expressed as follows:

$$X(Y) = (Y^2/R)/(1 + \sqrt{1 - (1+k) \times (Y/R)^2}) + \sum_i (Ai) \times (Y^i),$$

[0078] where,

[0079] X is the relative distance between a point on the aspheric surface spaced at a distance Y from the optical axis and the tangential plane at the aspheric surface vertex on the optical axis;

[0080] Y is the vertical distance from the point on the aspheric surface to the optical axis;

[0081] R is the curvature radius;

[0082] k is the conic coefficient; and

[0083] Ai is the i-th aspheric coefficient.

[0084] In the imaging system of the image capturing apparatus according to the 1st embodiment, when a focal length of the imaging system is f, an f-number of the imaging system is Fno, and half of a maximal field of view of the imaging system is HFOV, these parameters have the following values: f=4.26 mm; Fno=2.12; and HFOV=36.7 degrees.

[0085] In the imaging system of the image capturing apparatus according to the 1st embodiment, when an Abbe number of the second lens element 120 is V2, an Abbe number of the third lens element 130 is V3, and an Abbe number of the fourth lens element 140 is V4, the following conditions are satisfied: V2+V3+V4=77.2; (V2+V4)/V3=2.02; and V3=25.6.

[0086] In the imaging system of the image capturing apparatus according to the 1st embodiment, when a central thickness of the second lens element 120 is CT2, a central thickness of the third lens element 130 is CT3, a central thickness of the fourth lens element 140 is CT4, and a central thickness of the fifth lens element 150 is CT5, the following condition is satisfied: (CT2+CT3+CT4)/CT5=1.19.

[0087] In the imaging system of the image capturing apparatus according to the 1st embodiment, when an axial distance between the first lens element 110 and the second lens element 120 is T12, an axial distance between the second lens element 120 and the third lens element 130 is T23, an axial distance between the third lens element 130 and the fourth lens element 140 is T34, an axial distance between the fourth lens element 140 and the fifth lens element 150 is T45, and a sum of axial distances between any two of the first lens element 110, the second lens element 120, the third lens element 130, the fourth lens element 140 and the fifth lens element 150 that are adjacent to each other is ΣAT ($\Sigma AT=T12+T23+T34+T45$), the following condition is satisfied: (T34+T45)/ ΣAT =0.67.

[0088] In the imaging system of the image capturing apparatus according to the 1st embodiment, when the central thickness of the fourth lens element 140 is CT4, and the axial distance between the third lens element 130 and the fourth lens element 140 is T34, the following condition is satisfied: CT4/T34=0.75.

[0089] In the imaging system of the image capturing apparatus according to the 1st embodiment, when the central

thickness of the fourth lens element 140 is CT4, and the axial distance between the fourth lens element 140 and the fifth lens element 150 is T45, the following condition is satisfied: CT4/T45=0.85.

[0090] In the imaging system of the image capturing apparatus according to the 1st embodiment, when an axial distance between the object-side surface 111 of the first lens element 110 and the image surface 170 is TL, the following condition is satisfied: TL=4.79 mm.

[0091] In the imaging system of the image capturing apparatus according to the 1st embodiment, when the focal length of the imaging system is f, a curvature radius of the object-side surface 131 of the third lens element 130 is R5, and a curvature radius of the image-side surface 132 of the third lens element 130 is R6, the following condition is satisfied: |f/R5|+|f/R6|=0.30.

[0092] In the imaging system of the image capturing apparatus according to the 1st embodiment, when a curvature radius of the object-side surface 141 of the fourth lens element 140 is R7, and the focal length of the imaging system is f, the following condition is satisfied: |R7/f|=0.87.

[0093] In the imaging system of the image capturing apparatus according to the 1st embodiment, when the curvature radius of the object-side surface 141 of the fourth lens element 140 is R7, a curvature radius of the image-side surface 142 of the fourth lens element 140 is R8, a curvature radius of the object-side surface 151 of the fifth lens element 150 is R9, a curvature radius of the image-side surface 152 of the fifth lens element 150 is R10, and the focal length of the imaging system is f, the following condition is satisfied: (|R7|+|R8|+|R9|+|R10|)/f=2.81.

[0094] In the imaging system of the image capturing apparatus according to the 1st embodiment, when the curvature radius of the image-side surface 142 of the fourth lens element 140 is R8, and the curvature radius of the object-side surface 151 of the fifth lens element 150 is R9, the following condition is satisfied: R8/R9=1.39.

[0095] In the imaging system of the image capturing apparatus according to the 1st embodiment, when the focal length of the imaging system is f, a focal length of the third lens element 130 is f3, a focal length of the fourth lens element 140 is f4, and a focal length of the fifth lens element 150 is f5, the following condition is satisfied: |f/f3|+|f/f4|+|f/f5|=0.37.

[0096] Furthermore, in the imaging system of the image capturing apparatus according to the 1st embodiment, when a central thickness of the first lens element 110 is CT1, the central thickness of the second lens element 120 is CT2, the central thickness of the third lens element 130 is CT3, the central thickness of the fourth lens element 140 is CT4, and the central thickness of the fifth lens element 150 is CT5, CT5 is larger than CT1, CT2, CT3, and CT4.

[0097] The detailed optical data of the 1st embodiment are shown in TABLE 1 and the aspheric surface data are shown in TABLE 2 below.

TABLE 1

1st Embodiment								
$f = 4.26$ mm, $Fno = 2.12$, $HFOV = 36.7$ deg.								
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length	
0	Object	Plano	Infinity					
1	Ape. Stop	Plano	-0.393					
2	Lens 1	1.434 ASP	0.629	Plastic	1.535	55.7	3.00	
3		11.535 ASP	0.052					
4	Lens 2	16.839 ASP	0.230	Plastic	1.650	21.4	-6.97	
5		3.549 ASP	0.379					
6	Lens 3	-22.317 ASP	0.407	Plastic	1.614	25.6	-90.02	
7		-37.683 ASP	0.459					
8	Lens 4	3.714 ASP	0.344	Plastic	1.583	30.2	141.80	
9		3.756 ASP	0.403					
10	Lens 5	2.710 ASP	0.821	Plastic	1.530	55.8	-14.65	
11		1.798 ASP	0.500					
12	IR-cut filter	Plano	0.210	Glass	1.517	64.2	—	
13		Plano	0.355					
14	Image	Plano	—					

Note:

Reference wavelength is 587.6 nm (d-line).

TABLE 2

Aspheric Coefficients					
Surface #	2	3	4	5	6
k =	-3.9057E+00	8.1126E+01	-6.7700E+01	-2.7476E+01	8.4244E+01
A4 =	1.4800E-01	-2.0100E-01	-2.5701E-01	-3.7903E-02	-1.7204E-01
A6 =	3.1631E-02	4.0070E-01	7.1822E-01	4.4152E-01	3.0087E-02
A8 =	-2.1381E-01	-3.2339E-01	-8.1482E-01	-1.1901E+00	-5.1244E-02
A10 =	3.0089E-01	6.1536E-02	5.9678E-01	3.2337E+00	4.0113E-01
A12 =	-1.4879E-01	-1.1828E-03	-4.0758E-01	-5.6090E+00	-9.7297E-01
A14 =	-5.6929E-05	1.4320E-03	2.5424E-01	5.1544E+00	1.0740E+00
A16 =	-6.3564E-05	-3.9417E-04	-6.9494E-02	-1.8174E+00	-4.0524E-01
Surface #	7	8	9	10	11
k =	-9.0000E+01	-7.3247E+01	-7.1121E+01	-1.9266E+01	-6.5499E+00
A4 =	-1.4988E-01	4.5048E-02	-2.7074E-02	-1.5012E-01	-8.1795E-02
A6 =	-3.4820E-03	-2.2075E-01	-8.4917E-03	3.2235E-02	2.9107E-02
A8 =	1.3167E-01	2.5995E-01	-1.9701E-02	5.6233E-03	-1.1326E-02
A10 =	-1.9575E-01	-2.6800E-01	1.8416E-02	-2.8060E-03	3.3138E-03
A12 =	1.8094E-01	1.7767E-01	-6.4598E-03	2.9319E-04	-6.1702E-04
A14 =	-7.5692E-02	-6.6427E-02	1.1475E-03	5.1258E-06	6.2622E-05
A16 =	1.8109E-02	1.0581E-02	-9.7581E-05	-1.7248E-06	-2.5742E-06

[0098] In TABLE 1, the curvature radius, the thickness and the focal length are shown in millimeters (mm). Surface numbers 0-14 represent the surfaces sequentially arranged from the object-side to the image-side along the optical axis. In TABLE 2, k represents the conic coefficient of the equation of the aspheric surface profiles. A4-A16 represent the aspheric coefficients ranging from the 4th order to the 16th order. The tables presented below for each embodiment are the corresponding schematic parameter and aberration curves, and the definitions of the tables are the same as TABLE 1 and TABLE 2 of the 1st embodiment. Therefore, an explanation in this regard will not be provided again.

2nd Embodiment

[0099] FIG. 3 is a schematic view of an image capturing apparatus according to the 2nd embodiment of the present disclosure. FIG. 4 shows, in order from left to right, spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing apparatus according to the 2nd embodiment.

[0100] In FIG. 3, the image capturing apparatus includes the imaging system (its reference numeral is omitted) and an image sensor 280. The imaging system includes, in order from an object side to an image side, an aperture stop 200, a first lens element 210, a second lens element 220, a third lens element 230, a fourth lens element 240, a fifth lens element 250, an IR-cut filter 260 and an image surface 270. The image sensor 280 is disposed on the image surface 270 of the imaging system. The imaging system has a total of five lens elements (210-250) with refractive power. Moreover, there is an air gap between any two of the first lens element 210, the second lens element 220, the third lens element 230, the fourth lens element 240 and the fifth lens element 250 that are adjacent to each other.

[0101] The first lens element 210 with positive refractive power has an object-side surface 211 being convex in a paraxial region thereof and an image-side surface 212 being concave in a paraxial region thereof. The first lens element 210 is made of plastic material and has the object-side surface 211 and the image-side surface 212 being both aspheric.

[0102] The second lens element **220** with negative refractive power has an object-side surface **221** being convex in a paraxial region thereof and an image-side surface **222** being concave in a paraxial region thereof. The second lens element **220** is made of plastic material and has the object-side surface **221** and the image-side surface **222** being both aspheric.

[0103] The third lens element **230** with negative refractive power has an object-side surface **231** being concave in a paraxial region thereof and an image-side surface **232** being concave in a paraxial region thereof. The third lens element **230** is made of plastic material and has the object-side surface **231** and the image-side surface **232** being both aspheric.

[0104] The fourth lens element **240** with positive refractive power has an object-side surface **241** being convex in a paraxial region thereof and an image-side surface **242** being concave in a paraxial region thereof. The fourth lens element **240** is made of plastic material and has the object-side surface **241** and the image-side surface **242** being both aspheric. Furthermore, the image-side surface **242** of the fourth lens element **240** includes at least one convex shape in an off-axial region thereof.

[0105] The fifth lens element **250** with negative refractive power has an object-side surface **251** being convex in a

paraxial region thereof and an image-side surface **252** being concave in a paraxial region thereof. The fifth lens element **250** is made of plastic material and has the object-side surface **251** and the image-side surface **252** being both aspheric. Furthermore, the image-side surface **252** of the fifth lens element **250** includes at least one convex shape in an off-axial region thereof.

[0106] The IR-cut filter **260** is made of glass material and located between the fifth lens element **250** and the image surface **270**, and will not affect the focal length of the imaging system.

[0107] Furthermore, in the imaging system of the image capturing apparatus according to the 2nd embodiment, when a central thickness of the first lens element **210** is CT1, a central thickness of the second lens element **220** is CT2, a central thickness of the third lens element **230** is CT3, a central thickness of the fourth lens element **240** is CT4, and a central thickness of the fifth lens element **250** is CT5, CT5 is larger than CT1, CT2, CT3, and CT4.

[0108] The detailed optical data of the 2nd embodiment are shown in TABLE 3 and the aspheric surface data are shown in TABLE 4 below.

TABLE 3

2nd Embodiment								
f = 4.26 mm, Fno = 2.20, HFOV = 36.7 deg.								
Surface #		Curvature	Radius	Thickness	Material	Index	Abbe #	Focal Length
0	Object	Plano		Infinity				
1	Ape. Stop	Plano		-0.355				
2	Lens 1	1.433	ASP	0.598	Plastic	1.514	56.8	3.08
3		13.186	ASP	0.049				
4	Lens 2	14.070	ASP	0.230	Plastic	1.645	22.5	-6.75
5		3.301	ASP	0.418				
6	Lens 3	-29.974	ASP	0.310	Plastic	1.645	22.5	-39.96
7		184.090	ASP	0.455				
8	Lens 4	2.592	ASP	0.358	Plastic	1.583	30.2	142.22
9		2.539	ASP	0.482				
10	Lens 5	1.857	ASP	0.813	Plastic	1.514	56.8	-42.57
11		1.458	ASP	0.400				
12	IR-cut filter	Plano		0.210	Glass	1.517	64.2	—
13		Plano		0.502				
14	Image	Plano		—				

Note:

Reference wavelength is 587.6 nm (d-line).

TABLE 4

Aspheric Coefficients					
Surface #	2	3	4	5	6
k =	-3.2975E+00	8.1912E+01	8.7117E+01	-2.2073E+01	8.4244E+01
A4 =	1.2862E-01	-1.9618E-01	-2.5172E-01	-3.2237E-02	-2.2106E-01
A6 =	3.9649E-02	4.0625E-01	6.9076E-01	4.4486E-01	1.7191E-01
A8 =	-2.0150E-01	-3.3533E-01	-8.0504E-01	-1.2562E+00	-7.4596E-01
A10 =	2.9272E-01	5.6801E-02	5.8341E-01	3.3213E+00	2.2556E+00
A12 =	-1.5467E-01	5.4258E-03	-3.8992E-01	-5.6475E+00	-3.8055E+00
A14 =	3.7769E-04	4.4390E-03	2.5390E-01	5.1544E+00	3.3412E+00
A16 =	-6.3564E-05	-3.9417E-04	-6.9494E-02	-1.8174E+00	-1.1189E+00

TABLE 4-continued

Aspheric Coefficients					
Surface #	7	8	9	10	11
k =	-9.0000E+01	-6.0541E+01	-7.1121E+01	-1.8799E+01	-7.5324E+00
A4 =	-2.3927E-01	7.4805E-02	-1.5020E-02	-1.1421E-01	-5.3886E-02
A6 =	2.7268E-01	-1.9078E-01	2.8248E-02	1.1840E-02	1.1688E-02
A8 =	-7.2621E-01	1.7742E-01	-6.3923E-02	1.2119E-02	-3.3240E-03
A10 =	1.3911E+00	-1.5111E-01	4.4907E-02	-4.4270E-03	8.1285E-04
A12 =	-1.5388E+00	8.7045E-02	-1.5386E-02	5.6610E-04	-1.3714E-04
A14 =	9.2822E-01	-2.8469E-02	2.6362E-03	-1.9588E-05	1.2803E-05
A16 =	-2.1942E-01	3.9818E-03	-1.8365E-04	-8.2651E-07	-4.6805E-07

[0109] In the 2nd embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 2nd embodiment, so an explanation in this regard will not be provided again.

[0110] Moreover, these parameters can be calculated from TABLE 3 and TABLE 4 as the following values and satisfy the following conditions:

2nd Embodiment		
f (mm)	4.26 CT4/T34	0.79
Fno	2.20 CT4/T45	0.74
HFOV (deg.)	36.7 TL (mm)	4.83
V2 + V3 + V4	75.2 $ f/R5 + f/R6 $	0.17
(V2 + V4)/V3	2.34 $ R7/f $	0.61
V3	22.5 $(R7 + R8 + R9 + R10)/f$	1.98
(CT2 + CT3 + CT4)/CT5	1.10 R8/R9	1.37
(T34 + T45)/ΣAT	0.67 $ f/f3 + f/f4 + f/f5 $	0.24

3rd Embodiment

[0111] FIG. 5 is a schematic view of an image capturing apparatus according to the 3rd embodiment of the present disclosure. FIG. 6 shows, in order from left to right, spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing apparatus according to the 3rd embodiment.

[0112] In FIG. 5, the image capturing apparatus includes the imaging system (its reference numeral is omitted) and an image sensor 380. The imaging system includes, in order from an object side to an image side, an aperture stop 300, a first lens element 310, a second lens element 320, a third lens element 330, a fourth lens element 340, a fifth lens element 350, an IR-cut filter 360 and an image surface 370. The image sensor 380 is disposed on the image surface 370 of the imaging system. The imaging system has a total of five lens elements (310-350) with refractive power. Moreover, there is an air gap between any two of the first lens element 310, the second lens element 320, the third lens element 330, the fourth lens element 340 and the fifth lens element 350 that are adjacent to each other.

[0113] The first lens element 310 with positive refractive power has an object-side surface 311 being convex in a paraxial region thereof and an image-side surface 312 being convex in a paraxial region thereof. The first lens element 310 is made of plastic material and has the object-side surface 311 and the image-side surface 312 being both aspheric.

[0114] The second lens element 320 with negative refractive power has an object-side surface 321 being convex in a paraxial region thereof and an image-side surface 322 being concave in a paraxial region thereof. The second lens element 320 is made of plastic material and has the object-side surface 321 and the image-side surface 322 being both aspheric.

[0115] The third lens element 330 with positive power has an object-side surface 331 being convex in a paraxial region thereof and an image-side surface 332 being convex in a paraxial region thereof. The third lens element 330 is made of plastic material and has the object-side surface 331 and the image-side surface 332 being both aspheric.

[0116] The fourth lens element 340 with positive refractive power has an object-side surface 341 being convex in a paraxial region thereof and an image-side surface 342 being concave in a paraxial region thereof. The fourth lens element 340 is made of plastic material and has the object-side surface 341 and the image-side surface 342 being both aspheric. Furthermore, the image-side surface 342 of the fourth lens element 340 includes at least one convex shape in an off-axial region thereof.

[0117] The fifth lens element 350 with negative refractive power has an object-side surface 351 being convex in a paraxial region thereof and an image-side surface 352 being concave in a paraxial region thereof. The fifth lens element 350 is made of plastic material and has the object-side surface 351 and the image-side surface 352 being both aspheric. Furthermore, the image-side surface 352 of the fifth lens element 350 includes at least one convex shape in an off-axial region thereof.

[0118] The IR-cut filter 360 is made of glass material and located between the fifth lens element 350 and the image surface 370, and will not affect the focal length of the imaging system.

[0119] Furthermore, in the imaging system of the image capturing apparatus according to the 3rd embodiment, when a focal length of the first lens element 310 is $f1$, a focal length of the second lens element 320 is $f2$, a focal length of the third lens element 330 is $f3$, a focal length of the fourth lens element 340 is $f4$, and a focal length of the fifth lens element 350 is $f5$, the following conditions are satisfied: $|f3| > |f1|$; $|f3| > |f2|$; $|f3| > |f4|$; and $|f3| > |f5|$.

[0120] The detailed optical data of the 3rd embodiment are shown in TABLE 5 and the aspheric surface data are shown in TABLE 6 below.

TABLE 5

3rd Embodiment f = 3.88 mm, Fno = 2.05, HFOV = 37.0 deg.								
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length	
0	Object	Plano	Infinity					
1	Ape. Stop	Plano	-0.275					
2	Lens 1	1.681	ASP	0.766	Plastic	1.544	55.9	2.92
3		-24.398	ASP	0.085				
4	Lens 2	12.507	ASP	0.240	Plastic	1.640	23.3	-4.87
5		2.474	ASP	0.492				
6	Lens 3	13.266	ASP	0.340	Plastic	1.554	41.1	16.00
7		-26.498	ASP	0.485				
8	Lens 4	3.385	ASP	0.300	Plastic	1.530	56.0	6.60
9		100.000	ASP	0.442				
10	Lens 5	2.392	ASP	0.450	Plastic	1.544	55.9	-3.04
11		0.914	ASP	0.400				
12	IR-cut filter	Plano	0.110	Glass	1.517	64.2		—
13		Plano	0.280					
14	Image	Plano	—					

Note:

Reference wavelength is 587.6 nm (d-line).

TABLE 6

Aspheric Coefficients					
Surface #	2	3	4	5	6
k =	3.1504E-01	-9.0000E+01	-9.0000E+01	-2.3442E+01	2.5892E+01
A4 =	-2.1840E-02	-2.5550E-02	-6.5043E-02	1.1877E-01	-9.6721E-02
A6 =	4.1620E-02	1.4387E-01	2.1920E-01	-5.6213E-02	-2.5499E-01
A8 =	-1.2173E-01	-3.9128E-01	-4.5499E-01	1.9730E-03	7.8349E-01
A10 =	1.2637E-01	4.4331E-01	4.9730E-01	6.0775E-02	-1.2050E+00
A12 =	-5.4821E-02	-1.8355E-01	-1.9159E-01	-1.3858E-02	9.3455E-01
A14 =					-2.9675E-01
A16 =					2.1621E-02
Surface #	7	8	9	10	11
k =	8.9907E+01	7.0929E-01	6.5412E+01	-9.0000E+01	-9.7455E+00
A4 =	-1.2255E-01	1.6038E-01	1.2485E-01	-4.4319E-01	-1.6104E-01
A6 =	-1.4590E-01	-2.5373E-01	1.5214E-02	3.3110E-01	6.9374E-02
A8 =	1.8144E-01	2.0345E-01	-1.1281E-01	-1.3212E-01	-1.6428E-02
A10 =	1.2357E-01	-1.3552E-01	5.8076E-02	3.3089E-02	1.5595E-03
A12 =	-4.4768E-01	4.3918E-02	-1.1670E-02	-5.1641E-03	-2.2552E-05
A14 =	3.6866E-01	-5.0155E-03	8.4596E-04	4.5828E-04	5.7368E-06
A16 =	-9.5760E-02			-1.7642E-05	-1.0894E-06

[0121] In the 3rd embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 3rd embodiment, so an explanation in this regard will not be provided again.

[0122] Moreover, these parameters can be calculated from TABLE 5 and TABLE 6 as the following values and satisfy the following conditions:

3rd Embodiment			
f (mm)	3.88	CT4/T34	0.62
Fno	2.05	CT4/T45	0.68
HFOV (deg.)	37.0	TL (mm)	4.39
V2 + V3 + V4	120.4	f/R5 + f/R6	0.44
(V2 + V4)/V3	1.93	R7/f	0.87
V3	41.1	(R7 + R8 + R9 + R10)/f	27.50

-continued

3rd Embodiment			
(CT2 + CT3 + CT4)/CT5	1.96	R8/R9	41.81
(T34 + T45)/ΣAT	0.62	f/f3 + f/f4 + f/f5	2.11

4th Embodiment

[0123] FIG. 7 is a schematic view of an image capturing apparatus according to the 4th embodiment of the present disclosure. FIG. 8 shows, in order from left to right, spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing apparatus according to the 4th embodiment.

[0124] In FIG. 7, the image capturing apparatus includes the imaging system (its reference numeral is omitted) and an image sensor 480. The imaging system includes, in order from an object side to an image side, an aperture stop 400, a

first lens element **410**, a second lens element **420**, a third lens element **430**, a fourth lens element **440**, a fifth lens element **450**, an IR-cut filter **460** and an image surface **470**. The image sensor **480** is disposed on the image surface **470** of the imaging system. The imaging system has a total of five lens elements (**410-450**) with refractive power. Moreover, there is an air gap between any two of the first lens element **410**, the second lens element **420**, the third lens element **430**, the fourth lens element **440** and the fifth lens element **450** that are adjacent to each other.

[0125] The first lens element **410** with positive refractive power has an object-side surface **411** being convex in a paraxial region thereof and an image-side surface **412** being concave in a paraxial region thereof. The first lens element **410** is made of plastic material and has the object-side surface **411** and the image-side surface **412** being both aspheric.

[0126] The second lens element **420** with negative refractive power has an object-side surface **421** being convex in a paraxial region thereof and an image-side surface **422** being concave in a paraxial region thereof. The second lens element **420** is made of plastic material and has the object-side surface **421** and the image-side surface **422** being both aspheric.

[0127] The third lens element **430** with positive power has an object-side surface **431** being concave in a paraxial region thereof and an image-side surface **432** being convex in a paraxial region thereof. The third lens element **430** is made of plastic material and has the object-side surface **431** and the image-side surface **432** being both aspheric.

[0128] The fourth lens element **440** with negative refractive power has an object-side surface **441** being convex in a paraxial region thereof and an image-side surface **442** being concave in a paraxial region thereof. The fourth lens element **440** is made of plastic material and has the object-side surface **441** and the image-side surface **442** being both aspheric.

Furthermore, the image-side surface **442** of the fourth lens element **440** includes at least one convex shape in an off-axial region thereof.

[0129] The fifth lens element **450** with negative refractive power has an object-side surface **451** being convex in a paraxial region thereof and an image-side surface **452** being concave in a paraxial region thereof. The fifth lens element **450** is made of plastic material and has the object-side surface **451** and the image-side surface **452** being both aspheric. Furthermore, the image-side surface **452** of the fifth lens element **450** includes at least one convex shape in an off-axial region thereof.

[0130] The IR-cut filter **460** is made of glass material and located between the fifth lens element **450** and the image surface **470**, and will not affect the focal length of the imaging system.

[0131] Furthermore, in the imaging system of the image capturing apparatus according to the 4th embodiment, when a focal length of the first lens element **410** is f_1 , a focal length of the second lens element **420** is f_2 , a focal length of the third lens element **430** is f_3 , a focal length of the fourth lens element **440** is f_4 , and a focal length of the fifth lens element **450** is f_5 , the following conditions are satisfied: $|f_3| > |f_1|$; $|f_3| > |f_2|$; $|f_3| > |f_4|$; and $|f_3| > |f_5|$.

[0132] In the imaging system of the image capturing apparatus according to the 4th embodiment, when a central thickness of the first lens element **410** is CT_1 , a central thickness of the second lens element **420** is CT_2 , a central thickness of the third lens element **430** is CT_3 , a central thickness of the fourth lens element **440** is CT_4 , and a central thickness of the fifth lens element **450** is CT_5 , CT_5 is larger than CT_1 , CT_2 , CT_3 , and CT_4 .

[0133] The detailed optical data of the 4th embodiment are shown in TABLE 7 and the aspheric surface data are shown in TABLE 8 below.

TABLE 7

4th Embodiment								
$f = 4.20 \text{ mm}$, $Fno = 2.02$, $HFOV = 37.5 \text{ deg.}$								
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length	
0	Object	Plano	Infinity					
1	Ape. Stop	Plano	-0.417					
2	Lens 1	1.419 ASP	0.637	Plastic	1.544	55.9	2.88	
3		12.648 ASP	0.042					
4	Lens 2	42.726 ASP	0.231	Plastic	1.639	23.5	-6.52	
5		3.787 ASP	0.352					
6	Lens 3	-24.176 ASP	0.422	Plastic	1.639	23.5	319.05	
7		-21.760 ASP	0.478					
8	Lens 4	3.100 ASP	0.300	Plastic	1.639	23.5	-31.33	
9		2.583 ASP	0.425					
10	Lens 5	2.190 ASP	0.718	Plastic	1.535	55.7	-20.15	
11		1.612 ASP	0.400					
12	IR-cut filter	Plano	0.210	Glass	1.517	64.2	—	
13		Plano	0.423					
14	Image	Plano	—					

Note 1:

Reference wavelength is 587.6 nm (d-line).

Note 2:

Effective radius of Surface 7 is 1.000 mm.

TABLE 8

Aspheric Coefficients					
Surface #	2	3	4	5	6
k =	-3.9476E+00	8.9901E+01	-2.0380E+01	-2.8496E+01	3.4515E+01
A4 =	1.5297E-01	-2.1027E-01	-2.4650E-01	-2.2997E-02	-1.5694E-01
A6 =	3.1489E-02	4.0102E-01	6.6342E-01	3.2066E-01	9.5730E-02
A8 =	-2.1796E-01	-3.1333E-01	-6.6259E-01	-3.3200E-01	-3.1930E-01
A10 =	2.9932E-01	7.0404E-02	3.2112E-01	1.0412E-01	7.3215E-01
A12 =	-1.4745E-01	-5.8330E-03	-4.1919E-02	1.3084E-01	-7.9272E-01
A14 =					3.6953E-01
Surface #	7	8	9	10	11
k =	5.3708E+01	-8.9984E+01	-8.8257E+01	-2.9545E+01	-8.3226E+00
A4 =	-1.5089E-01	2.2868E-02	5.5897E-03	-1.5191E-01	-9.0796E-02
A6 =	1.3885E-01	-1.4897E-01	-9.3236E-02	2.6556E-02	3.3934E-02
A8 =	-2.9620E-01	4.7563E-02	6.8103E-02	1.1409E-02	-1.3919E-02
A10 =	4.7549E-01	3.0241E-02	-3.4918E-02	-5.0152E-03	4.1787E-03
A12 =	-3.7435E-01	-4.7233E-02	1.0767E-02	7.2775E-04	-7.8191E-04
A14 =	1.3433E-01	1.4791E-02	-1.3291E-03	-3.9272E-05	7.9620E-05
A16 =			-6.6516E-06	1.8507E-07	-3.2945E-06

[0134] In the 4th embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 4th embodiment, so an explanation in this regard will not be provided again.

[0135] Moreover, these parameters can be calculated from TABLE 7 and TABLE 8 as the following values and satisfy the following conditions:

4th Embodiment			
f (mm)	4.20	CT4/T34	0.63
Fno	2.02	CT4/T45	0.71
HFOV (deg.)	37.5	TL (mm)	4.64
V2 + V3 + V4	70.5	f/R5 + f/R6	0.37
(V2 + V4)/V3	2.00	R7/f	0.74
V3	23.5	(R7 + R8 + R9 + R10)/f	2.26
(CT2 + CT3 + CT4)/CT5	1.33	R8/R9	1.18
(T34 + T45)/ΣAT	0.70	f/f3 + f/f4 + f/f5	0.36

5th Embodiment

[0136] FIG. 9 is a schematic view of an image capturing apparatus according to the 5th embodiment of the present disclosure. FIG. 10 shows, in order from left to right, spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing apparatus according to the 5th embodiment.

[0137] In FIG. 9, the image capturing apparatus includes the imaging system (its reference numeral is omitted) and an image sensor 580. The imaging system includes, in order from an object side to an image side, an aperture stop 500, a first lens element 510, a second lens element 520, a third lens element 530, a fourth lens element 540, a fifth lens element 550, an IR-cut filter 560 and an image surface 570. The image sensor 580 is disposed on the image surface 570 of the imaging system. The imaging system has a total of five lens elements (510-550) with refractive power. Moreover, there is an air gap between any two of the first lens element 510, the

second lens element 520, the third lens element 530, the fourth lens element 540 and the fifth lens element 550 that are adjacent to each other.

[0138] The first lens element 510 with positive refractive power has an object-side surface 511 being convex in a paraxial region thereof and an image-side surface 512 being concave in a paraxial region thereof. The first lens element 510 is made of plastic material and has the object-side surface 511 and the image-side surface 512 being both aspheric.

[0139] The second lens element 520 with negative refractive power has an object-side surface 521 being convex in a paraxial region thereof and an image-side surface 522 being concave in a paraxial region thereof. The second lens element 520 is made of plastic material and has the object-side surface 521 and the image-side surface 522 being both aspheric.

[0140] The third lens element 530 with negative refractive power has an object-side surface 531 being concave in a paraxial region thereof and an image-side surface 532 being convex in a paraxial region thereof. The third lens element 530 is made of plastic material and has the object-side surface 531 and the image-side surface 532 being both aspheric.

[0141] The fourth lens element 540 with negative refractive power has an object-side surface 541 being convex in a paraxial region thereof and an image-side surface 542 being concave in a paraxial region thereof. The fourth lens element 540 is made of plastic material and has the object-side surface 541 and the image-side surface 542 being both aspheric. Furthermore, the image-side surface 542 of the fourth lens element 540 includes at least one convex shape in an off-axial region thereof.

[0142] The fifth lens element 550 with negative refractive power has an object-side surface 551 being convex in a paraxial region thereof and an image-side surface 552 being concave in a paraxial region thereof. The fifth lens element 550 is made of plastic material and has the object-side surface 551 and the image-side surface 552 being both aspheric. Furthermore, the image-side surface 552 of the fifth lens element 550 includes at least one convex shape in an off-axial region thereof.

[0143] The IR-cut filter **560** is made of glass material and located between the fifth lens element **550** and the image surface **570**, and will not affect the focal length of the imaging system.

[0144] Furthermore, in the imaging system of the image capturing apparatus according to the 5th embodiment, when a focal length of the first lens element **510** is f_1 , a focal length of the second lens element **520** is f_2 , a focal length of the third lens element **530** is f_3 , a focal length of the fourth lens element **540** is f_4 , and a focal length of the fifth lens element **550** is f_5 , the following conditions are satisfied: $|f_3| > |f_1|$; $|f_3| > |f_2|$; $|f_3| > |f_4|$; and $|f_3| > |f_5|$.

[0145] In the imaging system of the image capturing apparatus according to the 5th embodiment, when a central thickness of the first lens element **510** is CT_1 , a central thickness of the second lens element **520** is CT_2 , a central thickness of the third lens element **530** is CT_3 , a central thickness of the fourth lens element **540** is CT_4 , and a central thickness of the fifth lens element **550** is CT_5 , CT_5 is larger than CT_1 , CT_2 , CT_3 , and CT_4 .

[0146] The detailed optical data of the 5th embodiment are shown in TABLE 9 and the aspheric surface data are shown in TABLE 10 below.

TABLE 9

5th Embodiment							
$f = 4.19 \text{ mm}$, $Fno = 2.02$, $HFOV = 37.5 \text{ deg.}$							
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length
0	Object	Plano	Infinity				
1	Ape. Stop	Plano	-0.409				
2	Lens 1	1.422 ASP	0.628	Plastic	1.544	55.9	2.88
3		12.785 ASP	0.042				
4	Lens 2	65.061 ASP	0.230	Plastic	1.639	23.5	-6.56
5		3.932 ASP	0.347				
6	Lens 3	-19.301 ASP	0.402	Plastic	1.639	23.5	-2152.23
7		-19.735 ASP	0.474				
8	Lens 4	3.200 ASP	0.301	Plastic	1.639	23.5	-37.23
9		2.717 ASP	0.434				
10	Lens 5	2.082 ASP	0.713	Plastic	1.535	55.7	-21.81
11		1.556 ASP	0.400				
12	IR-cut filter	Plano	0.210	Glass	1.517	64.2	—
13		Plano	0.459				
14	Image	Plano	—				

Note 1:

Reference wavelength is 587.6 nm (d-line).

Note 2:

Effective radius of Surface 7 is 0.990 mm.

TABLE 10

Aspheric Coefficients					
Surface #	2	3	4	5	6
k =	-4.3410E+00	8.9901E+01	-2.0380E+01	-3.2967E+01	3.4515E+01
A4 =	1.7063E-01	-2.0149E-01	-2.2908E-01	-6.5327E-03	-1.5939E-01
A6 =	-1.6929E-03	3.8437E-01	6.3941E-01	2.7135E-01	6.4187E-02
A8 =	-1.7319E-01	-3.2421E-01	-6.7740E-01	-2.2748E-01	-1.8357E-01
A10 =	2.6395E-01	9.8898E-02	3.7293E-01	-3.1428E-02	4.5398E-01
A12 =	-1.3905E-01	-1.7931E-02	-6.5920E-02	2.1420E-01	-5.2698E-01
A14 =					2.9575E-01
Surface #	7	8	9	10	11
k =	5.3708E+01	-9.0000E+01	-8.9881E+01	-2.4765E+01	-7.6073E+00
A4 =	-1.4997E-01	3.1097E-02	6.1132E-04	-1.5404E-01	-9.8884E-02
A6 =	1.0273E-01	-1.7291E-01	-7.3771E-02	3.4391E-02	4.2029E-02
A8 =	-2.0582E-01	9.7627E-02	4.5126E-02	5.9047E-03	-1.7838E-02
A10 =	3.6118E-01	-2.8834E-02	-2.2341E-02	-3.2867E-03	5.2802E-03
A12 =	-3.0273E-01	-1.4635E-02	7.3166E-03	4.5032E-04	-9.5575E-04
A14 =	1.2149E-01	8.2738E-03	-8.2201E-04	-1.7484E-05	9.3717E-05
A16 =			-4.7331E-05	-4.5361E-07	-3.7484E-06

[0147] In the 5th embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 5th embodiment, so an explanation in this regard will not be provided again.

[0148] Moreover, these parameters can be calculated from TABLE 9 and TABLE 10 as the following values and satisfy the following conditions:

5th Embodiment			
f (mm)	4.19	CT4/T34	0.64
Fno	2.02	CT4/T45	0.69
HFOV (deg.)	37.5	TL (mm)	4.64
V2 + V3 + V4	70.5	f/R5 + f/R6	0.43
(V2 + V4)/V3	2.00	R7/f	0.76
V3	23.5	(R7 + R8 + R9 + R10)/f	2.28
(CT2 + CT3 + CT4)/CT5	1.31	R8/R9	1.31
(T34 + T45)/ΣAT	0.70	f/f3 + f/f4 + f/f5	0.31

6th Embodiment

[0149] FIG. 11 is a schematic view of an image capturing apparatus according to the 6th embodiment of the present disclosure. FIG. 12 shows, in order from left to right, spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing apparatus according to the 6th embodiment.

[0150] In FIG. 11, the image capturing apparatus includes the imaging system (its reference numeral is omitted) and an image sensor 680. The imaging system includes, in order from an object side to an image side, an aperture stop 600, a first lens element 610, a second lens element 620, a third lens element 630, a fourth lens element 640, a fifth lens element 650, an IR-cut filter 660 and an image surface 670. The image sensor 680 is disposed on the image surface 670 of the imaging system. The imaging system has a total of five lens elements (610-650) with refractive power. Moreover, there is an air gap between any two of the first lens element 610, the second lens element 620, the third lens element 630, the fourth lens element 640 and the fifth lens element 650 that are adjacent to each other.

[0151] The first lens element 610 with positive refractive power has an object-side surface 611 being convex in a paraxial region thereof and an image-side surface 612 being concave in a paraxial region thereof. The first lens element 610 is made of plastic material and has the object-side surface 611 and the image-side surface 612 being both aspheric.

[0152] The second lens element 620 with negative refractive power has an object-side surface 621 being concave in a paraxial region thereof and an image-side surface 622 being

concave in a paraxial region thereof. The second lens element 620 is made of plastic material and has the object-side surface 621 and the image-side surface 622 being both aspheric.

[0153] The third lens element 630 with negative refractive power has an object-side surface 631 being concave in a paraxial region thereof and an image-side surface 632 being convex in a paraxial region thereof. The third lens element 630 is made of plastic material and has the object-side surface 631 and the image-side surface 632 being both aspheric.

[0154] The fourth lens element 640 with negative refractive power has an object-side surface 641 being convex in a paraxial region thereof and an image-side surface 642 being concave in a paraxial region thereof. The fourth lens element 640 is made of plastic material and has the object-side surface 641 and the image-side surface 642 being both aspheric. Furthermore, the image-side surface 642 of the fourth lens element 640 includes at least one convex shape in an off-axial region thereof.

[0155] The fifth lens element 650 with positive refractive power has an object-side surface 651 being convex in a paraxial region thereof and an image-side surface 652 being concave in a paraxial region thereof. The fifth lens element 650 is made of plastic material and has the object-side surface 651 and the image-side surface 652 being both aspheric. Furthermore, the image-side surface 652 of the fifth lens element 650 includes at least one convex shape in an off-axial region thereof.

[0156] The IR-cut filter 660 is made of glass material and located between the fifth lens element 650 and the image surface 670, and will not affect the focal length of the imaging system.

[0157] Furthermore, in the imaging system of the image capturing apparatus according to the 6th embodiment, when a focal length of the first lens element 610 is f_1 , a focal length of the second lens element 620 is f_2 , a focal length of the third lens element 630 is f_3 , a focal length of the fourth lens element 640 is f_4 , and a focal length of the fifth lens element 650 is f_5 , the following conditions are satisfied: $|f_3| > |f_1|$; $|f_3| > |f_2|$; $|f_3| > |f_4|$; and $|f_3| > |f_5|$.

[0158] In the imaging system of the image capturing apparatus according to the 6th embodiment, when a central thickness of the first lens element 610 is CT_1 , a central thickness of the second lens element 620 is CT_2 , a central thickness of the third lens element 630 is CT_3 , a central thickness of the fourth lens element 640 is CT_4 , and a central thickness of the fifth lens element 650 is CT_5 , CT_5 is larger than CT_1 , CT_2 , CT_3 , and CT_4 .

[0159] The detailed optical data of the 6th embodiment are shown in TABLE 11 and the aspheric surface data are shown in TABLE 12 below.

TABLE 11

6th Embodiment							
f = 3.95 mm, Fno = 2.02, HFOV = 39.0 deg.							
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length
0	Object	Plano	Infinity				
1	Ape. Stop	Plano	-0.346				
2	Lens 1	1.426 ASP	0.579	Plastic	1.544	55.9	2.85
3		15.353 ASP	0.047				

TABLE 11-continued

6th Embodiment f = 3.95 mm, Fno = 2.02, HFOV = 39.0 deg.								
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length	
4	Lens 2	-19.744 ASP	0.230	Plastic	1.639	23.5	-6.47	
5		5.253 ASP	0.328					
6	Lens 3	-76.767 ASP	0.277	Plastic	1.639	23.5	-3284.25	
7		-79.794 ASP	0.490					
8	Lens 4	2.769 ASP	0.309	Plastic	1.639	23.5	-18.36	
9		2.143 ASP	0.390					
10	Lens 5	1.678 ASP	0.784	Plastic	1.535	55.7	74.22	
11		1.467 ASP	0.400					
12	IR-cut filter	Plano	0.210	Glass	1.517	64.2	—	
13		Plano	0.446					
14	Image	Plano	—					

Note 1:

Reference wavelength is 587.6 nm (d-line).

Note 2:

Effective radius of Surface 7 is 0.960 mm.

TABLE 12

Aspheric Coefficients					
Surface #	2	3	4	5	6
k =	-5.3333E+00	8.9974E+01	-2.0380E+01	-4.7812E+01	3.6034E+01
A4 =	2.0990E-01	-1.5946E-01	-1.5129E-01	-7.5717E-04	-2.0766E-01
A6 =	-9.1657E-02	2.7404E-01	5.0300E-01	2.5584E-01	-1.2692E-01
A8 =	-6.0971E-02	-1.9011E-01	-5.3516E-01	-3.0336E-01	6.1528E-01
A10 =	1.8782E-01	-2.4774E-02	2.7932E-01	1.1715E-01	-1.3369E+00
A12 =	-1.4150E-01	2.5520E-02	-3.4423E-02	8.4389E-02	1.4429E+00
A14 =					-4.7633E-01
Surface #	7	8	9	10	11
k =	5.3708E+01	-8.6375E+01	-6.3459E+01	-1.4849E+01	-5.5715E+00
A4 =	-2.1495E-01	4.5197E-02	-3.0688E-02	-1.7798E-01	-1.2090E-01
A6 =	1.2685E-01	-1.5434E-01	1.3038E-02	6.2494E-02	6.2092E-02
A8 =	-2.8129E-01	5.9273E-02	-5.1358E-02	-1.0083E-02	-2.6913E-02
A10 =	5.5123E-01	2.9224E-03	3.8419E-02	1.5161E-03	7.6184E-03
A12 =	-5.1325E-01	-2.7028E-02	-1.5065E-02	-3.4749E-04	-1.3038E-03
A14 =	2.5566E-01	9.9601E-03	3.5363E-03	5.2432E-05	1.2102E-04
A16 =			-3.8659E-04	-2.9969E-06	-4.6075E-06

[0160] In the 6th embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 6th embodiment, so an explanation in this regard will not be provided again.

[0161] Moreover, these parameters can be calculated from TABLE 11 and TABLE 12 as the following values and satisfy the following conditions:

6th Embodiment			
f (mm)	3.95	CT4/T34	0.63
Fno	2.02	CT4/T45	0.79
HFOV (deg.)	39.0	TL (mm)	4.49
V2 + V3 + V4	70.5	f/R5 + f/R6	0.10
(V2 + V4)/V3	2.00	R7/f	0.70
V3	23.5	(R7 + R8 + R9 + R10)/f	2.04
(CT2 + CT3 + CT4)/CT5	1.04	R8/R9	1.28
(T34 + T45)/ΣAT	0.70	f/f3 + f/f4 + f/f5	0.27

7th Embodiment

[0162] FIG. 13 is a schematic view of an image capturing apparatus according to the 7th embodiment of the present disclosure. FIG. 14 shows, in order from left to right, spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing apparatus according to the 7th embodiment.

[0163] In FIG. 13, the image capturing apparatus includes the imaging system (its reference numeral is omitted) and an image sensor 780. The imaging system includes, in order from an object side to an image side, a first lens element 710, an aperture stop 700, a second lens element 720, a third lens element 730, a fourth lens element 740, a fifth lens element 750, an IR-cut filter 760 and an image surface 770. The image sensor 780 is disposed on the image surface 770 of the imaging system. The imaging system has a total of five lens elements (710-750) with refractive power. Moreover, there is an air gap between any two of the first lens element 710, the second lens element 720, the third lens element 730, the fourth lens element 740 and the fifth lens element 750 that are adjacent to each other.

[0164] The first lens element **710** with positive refractive power has an object-side surface **711** being convex in a paraxial region thereof and an image-side surface **712** being convex in a paraxial region thereof. The first lens element **710** is made of plastic material and has the object-side surface **711** and the image-side surface **712** being both aspheric.

[0165] The second lens element **720** with negative refractive power has an object-side surface **721** being concave in a paraxial region thereof and an image-side surface **722** being concave in a paraxial region thereof. The second lens element **720** is made of plastic material and has the object-side surface **721** and the image-side surface **722** being both aspheric.

[0166] The third lens element **730** with positive refractive power has an object-side surface **731** being convex in a paraxial region thereof and an image-side surface **732** being convex in a paraxial region thereof. The third lens element **730** is made of plastic material and has the object-side surface **731** and the image-side surface **732** being both aspheric.

[0167] The fourth lens element **740** with negative refractive power has an object-side surface **741** being convex in a paraxial region thereof and an image-side surface **742** being concave in a paraxial region thereof. The fourth lens element **740** is made of plastic material and has the object-side surface **741** and the image-side surface **742** being both aspheric.

Furthermore, the image-side surface **742** of the fourth lens element **740** includes at least one convex shape in an off-axial region thereof.

[0168] The fifth lens element **750** with positive refractive power has an object-side surface **751** being convex in a paraxial region thereof and an image-side surface **752** being concave in a paraxial region thereof. The fifth lens element **750** is made of plastic material and has the object-side surface **751** and the image-side surface **752** being both aspheric. Furthermore, the image-side surface **752** of the fifth lens element **750** includes at least one convex shape in an off-axial region thereof.

[0169] The IR-cut filter **760** is made of glass material and located between the fifth lens element **750** and the image surface **770**, and will not affect the focal length of the imaging system.

[0170] Furthermore, in the imaging system of the image capturing apparatus according to the 7th embodiment, when a central thickness of the first lens element **710** is CT1, a central thickness of the second lens element **720** is CT2, a central thickness of the third lens element **730** is CT3, a central thickness of the fourth lens element **740** is CT4, and a central thickness of the fifth lens element **750** is CT5, CT5 is larger than CT1, CT2, CT3, and CT4.

[0171] The detailed optical data of the 7th embodiment are shown in TABLE 13 and the aspheric surface data are shown in TABLE 14 below.

TABLE 13

7th Embodiment								
f = 4.28 mm, Fno = 2.35, HFOV = 36.8 deg.								
Surface #		Curvature	Radius	Thickness	Material	Index	Abbe #	Focal Length
0	Object	Plano		Infinity				
1	Lens 1	1.566	ASP	0.595	Plastic	1.544	55.9	2.83
2		-90.733	ASP	0.002				
3	Ape. Stop	Plano		0.040				
4	Lens 2	-114.780	ASP	0.245	Plastic	1.639	23.5	-5.59
5		3.692	ASP	0.347				
6	Lens 3	97.739	ASP	0.425	Plastic	1.639	23.5	34.77
7		-28.710	ASP	0.593				
8	Lens 4	5.042	ASP	0.458	Plastic	1.639	23.5	-11.78
9		2.913	ASP	0.314				
10	Lens 5	1.929	ASP	0.809	Plastic	1.544	55.9	191.52
11		1.675	ASP	0.400				
12	IR-cut filter	Plano		0.300	Glass	1.517	64.2	—
13		Plano		0.381				
14	Image	Plano		—				

Note:

Reference wavelength is 587.6 nm (d-line).

TABLE 14

Aspheric Coefficients					
Surface #	1	2	4	5	6
k =	-3.3259E+00	-9.0000E+01	-9.0000E+01	-3.2984E+01	8.9864E+01
A4 =	1.0545E-01	-1.7243E-01	-2.1888E-01	-4.1211E-02	-1.6732E-01
A6 =	-5.2389E-02	4.1323E-01	6.2855E-01	3.9320E-01	1.5354E-01
A8 =	9.1540E-02	-4.1649E-01	-4.0350E-01	-1.1232E+00	-5.5628E-01
A10 =	-1.8423E-01	3.0140E-01	-7.7626E-01	2.8071E+00	1.4735E+00
A12 =	1.6136E-01	-4.7460E-01	2.0824E+00	-4.6165E+00	-2.3056E+00
A14 =	-5.1759E-02	5.7776E-01	-1.9367E+00	4.1896E+00	2.0069E+00
A16 =	-1.0472E-02	-2.6979E-01	6.6968E-01	-1.5715E+00	-7.2635E-01

TABLE 14-continued

Aspheric Coefficients					
Surface #	7	8	9	10	11
k =	-2.6176E+01	-9.0000E+01	-7.3862E+01	-2.2539E+01	-9.4754E+00
A4 =	-1.2240E-01	-3.5331E-02	7.8637E-04	-8.2850E-02	-4.4571E-02
A6 =	3.4599E-02	9.1346E-03	-3.2783E-02	3.3239E-03	9.3277E-03
A8 =	2.7087E-02	-1.0518E-01	8.1468E-03	7.1674E-03	-3.2288E-03
A10 =	-9.2145E-02	1.1676E-01	2.6828E-03	-1.7762E-03	8.0066E-04
A12 =	1.4832E-01	-6.5693E-02	-2.3517E-03	1.3682E-04	-1.2446E-04
A14 =	-9.0621E-02	1.6391E-02	6.2468E-04	2.8413E-06	1.0987E-05
A16 =	2.6153E-02	-1.1273E-03	-5.9896E-05	-6.4000E-07	-4.0222E-07

[0172] In the 7th embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 7th embodiment, so an explanation in this regard will not be provided again.

[0173] Moreover, these parameters can be calculated from TABLE 13 and TABLE 14 as the following values and satisfy the following conditions:

7th Embodiment			
f (mm)	4.28	CT4/T34	0.77
Fno	2.35	CT4/T45	1.46
HFOV (deg.)	36.8	TL (mm)	4.91
V2 + V3 + V4	70.5	f/R5 + f/R6	0.19
(V2 + V4)/V3	2.00	R7/f	1.18
V3	23.5	(R7 + R8 + R9 + R10)/f	2.70
(CT2 + CT3 + CT4)/CT5	1.39	R8/R9	1.51
(T34 + T45)/ΣAT	0.70	f/f3 + f/f4 + f/f5	0.51

8th Embodiment

[0174] FIG. 15 is a schematic view of an image capturing apparatus according to the 8th embodiment of the present disclosure. FIG. 16 shows, in order from left to right, spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing apparatus according to the 8th embodiment.

[0175] In FIG. 15, the image capturing apparatus includes the imaging system (its reference numeral is omitted) and an image sensor 880. The imaging system includes, in order from an object side to an image side, a first lens element 810, an aperture stop 800, a second lens element 820, a third lens element 830, a fourth lens element 840, a fifth lens element 850, an IR-cut filter 860 and an image surface 870. The image sensor 880 is disposed on the image surface 870 of the imaging system. The imaging system has a total of five lens elements (810-850) with refractive power. Moreover, there is an air gap between any two of the first lens element 810, the second lens element 820, the third lens element 830, the fourth lens element 840 and the fifth lens element 850 that are adjacent to each other.

[0176] The first lens element 810 with positive refractive power has an object-side surface 811 being convex in a paraxial region thereof and an image-side surface 812 being concave in a paraxial region thereof. The first lens element 810 is made of plastic material and has the object-side surface 811 and the image-side surface 812 being both aspheric.

[0177] The second lens element 820 with negative refractive power has an object-side surface 821 being concave in a paraxial region thereof and an image-side surface 822 being concave in a paraxial region thereof. The second lens element 820 is made of plastic material and has the object-side surface 821 and the image-side surface 822 being both aspheric.

[0178] The third lens element 830 with positive refractive power has an object-side surface 831 being convex in a paraxial region thereof and an image-side surface 832 being convex in a paraxial region thereof. The third lens element 830 is made of plastic material and has the object-side surface 831 and the image-side surface 832 being both aspheric.

[0179] The fourth lens element 840 with negative refractive power has an object-side surface 841 being convex in a paraxial region thereof and an image-side surface 842 being concave in a paraxial region thereof. The fourth lens element 840 is made of plastic material and has the object-side surface 841 and the image-side surface 842 being both aspheric. Furthermore, the image-side surface 842 of the fourth lens element 840 includes at least one convex shape in an off-axial region thereof.

[0180] The fifth lens element 850 with positive refractive power has an object-side surface 851 being convex in a paraxial region thereof and an image-side surface 852 being concave in a paraxial region thereof. The fifth lens element 850 is made of plastic material and has the object-side surface 851 and the image-side surface 852 being both aspheric. Furthermore, the image-side surface 852 of the fifth lens element 850 includes at least one convex shape in an off-axial region thereof.

[0181] The IR-cut filter 860 is made of glass material and located between the fifth lens element 850 and the image surface 870, and will not affect the focal length of the imaging system.

[0182] Furthermore, in the imaging system of the image capturing apparatus according to the 8th embodiment, when a central thickness of the first lens element 810 is CT1, a central thickness of the second lens element 820 is CT2, a central thickness of the third lens element 830 is CT3, a central thickness of the fourth lens element 840 is CT4, and a central thickness of the fifth lens element 850 is CT5, CT5 is larger than CT1, CT2, CT3, and CT4.

[0183] The detailed optical data of the 8th embodiment are shown in TABLE 15 and the aspheric surface data are shown in TABLE 16 below.

TABLE 15

8th Embodiment f = 4.05 mm, Fno = 2.35, HFOV = 39.5 deg.								
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length	
0	Object	Plano	Infinity					
1	Lens 1	1.467	ASP	0.516	Plastic	1.544	55.9	2.92
2		16.705	ASP	-0.001				
3	Ape. Stop	Plano	0.046					
4	Lens 2	-30.202	ASP	0.230	Plastic	1.639	23.5	-5.35
5		3.865	ASP	0.275				
6	Lens 3	21.016	ASP	0.353	Plastic	1.639	23.5	16.32
7		-20.570	ASP	0.634				
8	Lens 4	4.434	ASP	0.434	Plastic	1.639	23.5	-10.32
9		2.550	ASP	0.310				
10	Lens 5	1.616	ASP	0.790	Plastic	1.544	55.9	27.71
11		1.499	ASP	0.400				
12	IR-cut filter	Plano	0.300	Glass	1.517	64.2	—	
13		Plano	0.403					
14	Image	Plano	—					

Note:

Reference wavelength is 587.6 nm (d-line).

TABLE 16

Aspheric Coefficients					
Surface #	1	2	4	5	6
k =	-2.7176E+00	-9.0000E+01	-9.0000E+01	-3.0690E+01	9.0000E+01
A4 =	1.0448E-01	-1.8731E-01	-1.7331E-01	-3.9935E-03	-1.9372E-01
A6 =	-5.7505E-02	4.0362E-01	6.3574E-01	4.6459E-01	1.0723E-01
A8 =	8.3519E-02	-4.2358E-01	-4.0759E-01	-1.2196E+00	-3.7569E-01
A10 =	-1.9767E-01	2.9514E-01	-7.7476E-01	3.0670E+00	9.6879E-01
A12 =	1.4725E-01	-4.8274E-01	2.1229E+00	-5.4701E+00	-1.6133E+00
A14 =	-6.2689E-02	5.6560E-01	-1.9259E+00	5.6807E+00	1.5718E+00
A16 =	-1.6835E-02	-2.8763E-01	6.0436E-01	-2.4295E+00	-5.3975E-01
Surface #	7	8	9	10	11
k =	-2.6176E+01	-3.0443E+01	-5.9043E+01	-1.7769E+01	-8.5133E+00
A4 =	-1.3359E-01	-3.4965E-02	3.4456E-02	-7.7629E-02	-4.4041E-02
A6 =	-1.0344E-02	2.5686E-02	-5.9695E-02	3.0829E-03	9.0785E-03
A8 =	1.5628E-01	-1.1966E-01	2.9831E-02	7.0914E-03	-3.2398E-03
A10 =	-4.2840E-01	1.3477E-01	-1.0620E-02	-1.7832E-03	8.0156E-04
A12 =	6.8465E-01	-8.4816E-02	2.7247E-03	1.3672E-04	-1.2422E-04
A14 =	-5.2614E-01	2.7249E-02	-4.0503E-04	2.9414E-06	1.1017E-05
A16 =	1.8792E-01	-3.3318E-03	2.4167E-05	-6.0969E-07	-4.0003E-07

[0184] In the 8th embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 8th embodiment, so an explanation in this regard will not be provided again.

[0185] Moreover, these parameters can be calculated from TABLE 15 and TABLE 16 as the following values and satisfy the following conditions:

8th Embodiment			
f (mm)	4.05	CT4/T34	0.68
Fno	2.35	CT4/T45	1.40
HFOV (deg.)	39.5	TL (mm)	4.69
V2 + V3 + V4	70.5	f/R5 + f/R6	0.39
(V2 + V4)/V3	2.00	R7/f	1.09
V3	23.5	(R7 + R8 + R9 + R10)/f	2.49

-continued

8th Embodiment			
(CT2 + CT3 + CT4)/CT5	1.29	R8/R9	1.58
(T34 + T45)/ΣAT	0.75	f/f3 + f/f4 + f/f5	0.79

9th Embodiment

[0186] FIG. 17 is a schematic view of an image capturing apparatus according to the 9th embodiment of the present disclosure. FIG. 18 shows, in order from left to right, spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing apparatus according to the 9th embodiment.

[0187] In FIG. 17, the image capturing apparatus includes the imaging system (its reference numeral is omitted) and an image sensor 980. The imaging system includes, in order from an object side to an image side, an aperture stop 900, a

first lens element **910**, a second lens element **920**, a third lens element **930**, a fourth lens element **940**, a fifth lens element **950**, an IR-cut filter **960** and an image surface **970**. The image sensor **980** is disposed on the image surface **970** of the imaging system. The imaging system has a total of five lens elements (**910-950**) with refractive power. Moreover, there is an air gap between any two of the first lens element **910**, the second lens element **920**, the third lens element **930**, the fourth lens element **940** and the fifth lens element **950** that are adjacent to each other.

[0188] The first lens element **910** with positive refractive power has an object-side surface **911** being convex in a paraxial region thereof and an image-side surface **912** being concave in a paraxial region thereof. The first lens element **910** is made of plastic material and has the object-side surface **911** and the image-side surface **912** being both aspheric.

[0189] The second lens element **920** with negative refractive power has an object-side surface **921** being convex in a paraxial region thereof and an image-side surface **922** being concave in a paraxial region thereof. The second lens element **920** is made of plastic material and has the object-side surface **921** and the image-side surface **922** being both aspheric.

[0190] The third lens element **930** with positive refractive power has an object-side surface **931** being convex in a paraxial region thereof and an image-side surface **932** being concave in a paraxial region thereof. The third lens element **930** is made of plastic material and has the object-side surface **931** and the image-side surface **932** being both aspheric.

[0191] The fourth lens element **940** with positive refractive power has an object-side surface **941** being convex in a paraxial region thereof and an image-side surface **942** being concave in a paraxial region thereof. The fourth lens element **940** is made of plastic material and has the object-side surface **941** and the image-side surface **942** being both aspheric.

Furthermore, the image-side surface **942** of the fourth lens element **940** includes at least one convex shape in an off-axial region thereof.

[0192] The fifth lens element **950** with positive refractive power has an object-side surface **951** being convex in a paraxial region thereof and an image-side surface **952** being concave in a paraxial region thereof. The fifth lens element **950** is made of plastic material and has the object-side surface **951** and the image-side surface **952** being both aspheric. Furthermore, the image-side surface **952** of the fifth lens element **950** includes at least one convex shape in an off-axial region thereof.

[0193] The IR-cut filter **960** is made of glass material and located between the fifth lens element **950** and the image surface **970**, and will not affect the focal length of the imaging system.

[0194] Furthermore, in the imaging system of the image capturing apparatus according to the 9th embodiment, when a focal length of the first lens element **910** is f_1 , a focal length of the second lens element **920** is f_2 , a focal length of the third lens element **930** is f_3 , a focal length of the fourth lens element **940** is f_4 , and a focal length of the fifth lens element **950** is f_5 , the following conditions are satisfied: $|f_3| > |f_1|$; $|f_3| > |f_2|$; $|f_3| > |f_4|$; and $|f_3| > |f_5|$.

[0195] In the imaging system of the image capturing apparatus according to the 9th embodiment, when a central thickness of the first lens element **910** is CT_1 , a central thickness of the second lens element **920** is CT_2 , a central thickness of the third lens element **930** is CT_3 , a central thickness of the fourth lens element **940** is CT_4 , and a central thickness of the fifth lens element **950** is CT_5 , CT_5 is larger than CT_1 , CT_2 , CT_3 , and CT_4 .

[0196] The detailed optical data of the 9th embodiment are shown in TABLE 17 and the aspheric surface data are shown in TABLE 18 below.

TABLE 17

9th Embodiment								
$f = 4.04 \text{ mm}$, $Fno = 2.20$, $HFOV = 38.2 \text{ deg.}$								
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length	
0	Object	Plano	Infinity					
1	Ape. Stop	Plano	-0.327					
2	Lens 1	1.452 ASP	0.515	Plastic	1.535	55.7	3.05	
3		11.605 ASP	0.054					
4	Lens 2	9.560 ASP	0.240	Plastic	1.639	23.5	-5.06	
5		2.391 ASP	0.339					
6	Lens 3	7.860 ASP	0.260	Plastic	1.607	26.6	96.01	
7		8.970 ASP	0.424					
8	Lens 4	2.422 ASP	0.332	Plastic	1.583	30.2	65.97	
9		2.454 ASP	0.555					
10	Lens 5	1.567 ASP	0.762	Plastic	1.514	56.8	38.44	
11		1.421 ASP	0.400					
12	IR-cut filter	Plano	0.210	Glass	1.517	64.2	—	
13		Plano	0.601					
14	Image	Plano	—					

Note:

Reference wavelength is 587.6 nm (d-line).

TABLE 18

Aspheric Coefficients					
Surface #	2	3	4	5	6
k =	-3.1911E+00	9.0000E+01	1.5426E+01	-1.4164E+01	2.2746E+01
A4 =	1.2148E-01	-1.9127E-01	-2.6066E-01	-1.4643E-03	-2.6357E-01
A6 =	3.8839E-02	4.0471E-01	7.0067E-01	4.6136E-01	4.6096E-01
A8 =	-1.9616E-01	-3.3644E-01	-7.8040E-01	-1.2949E+00	-1.7920E+00
A10 =	2.9560E-01	6.0338E-02	5.3976E-01	3.3250E+00	4.6139E+00
A12 =	-1.5274E-01	1.4423E-02	-5.0192E-01	-5.6068E+00	-6.8958E+00
A14 =	2.4472E-03	7.4058E-03	4.3887E-01	5.0607E+00	5.5061E+00
A16 =	-3.0268E-03	-1.6242E-02	-1.5310E-01	-1.8012E+00	-1.7519E+00
Surface #	7	8	9	10	11
k =	-8.9680E+01	-5.6522E+01	-5.6903E+01	-9.3170E+00	-5.4248E+00
A4 =	-2.7532E-01	7.6743E-02	-4.5528E-02	-9.1950E-02	-5.9617E-02
A6 =	5.2607E-01	-1.6757E-01	1.1855E-01	3.8924E-03	1.3412E-02
A8 =	-1.5836E+00	1.6565E-01	-1.5783E-01	1.1789E-02	-3.2822E-03
A10 =	2.9676E+00	-1.6726E-01	9.3850E-02	-4.2375E-03	5.2692E-04
A12 =	-3.2121E+00	9.9890E-02	-2.9134E-02	6.8265E-04	-4.2249E-05
A14 =	1.8714E+00	-3.0189E-02	4.6521E-03	-5.5525E-05	7.1991E-07
A16 =	-4.2724E-01	3.7226E-03	-3.1116E-04	1.8696E-06	8.4716E-08

[0197] In the 9th embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 9th embodiment, so an explanation in this regard will not be provided again.

[0198] Moreover, these parameters can be calculated from TABLE 17 and TABLE 18 as the following values and satisfy the following conditions:

9th Embodiment			
f (mm)	4.04	CT4/T34	0.78
Fno	2.20	CT4/T45	0.60
HFOV (deg.)	38.2	TL (mm)	4.69
V2 + V3 + V4	80.3	f/R5 + f/R6	0.96
(V2 + V4)/V3	2.02	R7/f	0.60
V3	26.6	(R7 + R8 + R9 + R10)/f	1.95
(CT2 + CT3 + CT4)/CT5	1.09	R8/R9	1.57
(T34 + T45)/ΣAT	0.71	f/f3 + f/f4 + f/f5	0.21

10th Embodiment

[0199] FIG. 19 is a schematic view of an image capturing apparatus according to the 10th embodiment of the present disclosure. FIG. 20 shows, in order from left to right, spherical aberration curves, astigmatic field curves and a distortion curve of the image capturing apparatus according to the 10th embodiment.

[0200] In FIG. 19, the image capturing apparatus includes the imaging system (its reference numeral is omitted) and an image sensor 1080. The imaging system includes, in order from an object side to an image side, an aperture stop 1000, a first lens element 1010, a second lens element 1020, a third lens element 1030, a fourth lens element 1040, a fifth lens element 1050, an IR-cut filter 1060 and an image surface 1070. The image sensor 180 is disposed on the image surface 1070 of the imaging system. The imaging system has a total of five lens elements (1010-1050) with refractive power. Moreover, there is an air gap between any two of the first lens element 1010, the second lens element 1020, the third lens

element 1030, the fourth lens element 1040 and the fifth lens element 1050 that are adjacent to each other.

[0201] The first lens element 1010 with positive refractive power has an object-side surface 1011 being convex in a paraxial region thereof and an image-side surface 1012 being concave in a paraxial region thereof. The first lens element 1010 is made of plastic material and has the object-side surface 1011 and the image-side surface 1012 being both aspheric.

[0202] The second lens element 1020 with negative refractive power has an object-side surface 1021 being convex in a paraxial region thereof and an image-side surface 1022 being concave in a paraxial region thereof. The second lens element 1020 is made of plastic material and has the object-side surface 1021 and the image-side surface 1022 being both aspheric.

[0203] The third lens element 1030 with positive refractive power has an object-side surface 1031 being convex in a paraxial region thereof and an image-side surface 1032 being concave in a paraxial region thereof. The third lens element 1030 is made of plastic material and has the object-side surface 1031 and the image-side surface 1032 being both aspheric.

[0204] The fourth lens element 1040 with negative refractive power has an object-side surface 1041 being convex in a paraxial region thereof and an image-side surface 1042 being concave in a paraxial region thereof. The fourth lens element 1040 is made of plastic material and has the object-side surface 1041 and the image-side surface 1042 being both aspheric. Furthermore, the image-side surface 1042 of the fourth lens element 1040 includes at least one convex shape in an off-axial region thereof.

[0205] The fifth lens element 1050 with negative refractive power has an object-side surface 1051 being convex in a paraxial region thereof and an image-side surface 1052 being concave in a paraxial region thereof. The fifth lens element 1050 is made of plastic material and has the object-side surface 1051 and the image-side surface 1052 being both aspheric. Furthermore, the image-side surface 1052 of the fifth lens element 1050 includes at least one convex shape in an off-axial region thereof.

[0206] The IR-cut filter **1060** is made of glass material and located between the fifth lens element **1050** and the image surface **1070**, and will not affect the focal length of the imaging system.

[0207] Furthermore, in the imaging system of the image capturing apparatus according to the 10th embodiment, when a central thickness of the first lens element **1010** is CT1, a

central thickness of the second lens element **1020** is CT2, a central thickness of the third lens element **1030** is CT3, a central thickness of the fourth lens element **1040** is CT4, and a central thickness of the fifth lens element **1050** is CT5, CT5 is larger than CT1, CT2, CT3, and CT4.

[0208] The detailed optical data of the 10th embodiment are shown in TABLE 19 and the aspheric surface data are shown in TABLE 20 below.

TABLE 19

10th Embodiment								
f = 4.12 mm, Fno = 2.02, HFOV = 37.8 deg.								
Surface #		Curvature Radius	Thickness	Material	Index	Abbe #	Focal Length	
0	Object	Plano	Infinity					
1	Ape. Stop	Plano	−0.374					
2	Lens 1	1.440 ASP	0.606	Plastic	1.544	55.9	3.03	
3		9.742 ASP	0.035					
4	Lens 2	10.675 ASP	0.254	Plastic	1.639	23.5	−6.38	
5		2.922 ASP	0.371					
6	Lens 3	47.774 ASP	0.349	Plastic	1.639	23.5	38.90	
7		−51.679 ASP	0.447					
8	Lens 4	3.068 ASP	0.300	Plastic	1.639	23.5	−19.13	
9		2.359 ASP	0.381					
10	Lens 5	1.960 ASP	0.802	Plastic	1.535	55.7	−187.10	
11		1.648 ASP	0.400					
12	IR-cut filter	Plano	0.210	Glass	1.517	64.2	—	
13		Plano	0.489					
14	Image	Plano	—					

Note 1:

Reference wavelength is 587.6 nm (d-line).

Note 2:

Effective radius of Surface 7 is 1.030 mm.

TABLE 20

Aspheric Coefficients					
Surface #	2	3	4	5	6
k =	−3.9637E+00	−4.0773E+01	−9.0000E+01	−8.1984E+00	9.0000E+01
A4 =	1.4445E−01	−2.4003E−01	−2.3866E−01	2.6847E−03	−1.2383E−01
A6 =	1.7143E−02	4.3206E−01	5.6981E−01	2.8728E−01	2.3562E−02
A8 =	−1.9236E−01	−2.5951E−01	−3.4169E−01	−2.7760E−01	8.3624E−02
A10 =	2.5762E−01	−5.8016E−02	−6.5144E−02	2.2817E−01	−2.2683E−01
A12 =	−1.3296E−01	5.5852E−02	1.0363E−01	−4.9720E−02	3.0188E−01
A14 =					−1.0878E−01
Surface #	7	8	9	10	11
k =	9.0000E+01	−9.0000E+01	−6.1488E+00	−1.6210E+01	−5.9694E+00
A4 =	−1.3961E−01	5.3575E−03	−5.7094E−02	−1.9193E−01	−1.1218E−01
A6 =	5.9318E−02	−8.4418E−02	4.2169E−02	7.3749E−02	4.9904E−02
A8 =	−6.5968E−02	4.8769E−03	−4.0776E−02	−8.5671E−03	−1.8098E−02
A10 =	7.5868E−02	3.3853E−02	1.0251E−02	−1.2736E−03	4.2563E−03
A12 =	−4.5481E−02	−4.2129E−02	2.2066E−03	4.8807E−04	−6.1120E−04
A14 =	2.7783E−02	1.3411E−02	−1.2266E−03	−5.3467E−05	4.8991E−05
A16 =			1.2529E−04	2.0729E−06	−1.6637E−06

[0209] In the 10th embodiment, the equation of the aspheric surface profiles of the aforementioned lens elements is the same as the equation of the 1st embodiment. Also, the definitions of these parameters shown in the following table are the same as those stated in the 1st embodiment with corresponding values for the 10th embodiment, so an explanation in this regard will not be provided again.

[0210] Moreover, these parameters can be calculated from TABLE 19 and TABLE 20 as the following values and satisfy the following conditions:

10th Embodiment			
f (mm)	4.12	CT4/T34	0.67
Fno	2.02	CT4/T45	0.79
HFOV (deg.)	37.8	TL (mm)	4.64
V2 + V3 + V4	70.5	f/R5 + f/R6	0.17
(V2 + V4)/V3	2.00	R7/f	0.74
V3	23.5	(R7 + R8 + R9 + R10)/f	2.19
(CT2 + CT3 + CT4)/CT5	1.13	R8/R9	1.20
(T34 + T45)/ΣAT	0.67	f/f3 + f/f4 + f/f5	0.34

11th Embodiment

[0211] FIG. 21 is a schematic view of an electronic device 10 according to the 11th embodiment of the present disclosure. The electronic device 10 of the 11th embodiment is a smart phone, wherein the electronic device 10 includes an image capturing apparatus 11. The image capturing apparatus 11 includes an imaging system (its reference numeral is omitted) according to the present disclosure and an image sensor (its reference numeral is omitted), wherein the image sensor is disposed on an image surface of the imaging system.

12th Embodiment

[0212] FIG. 22 is a schematic view of an electronic device 20 according to the 12th embodiment of the present disclosure. The electronic device 20 of the 12th embodiment is a tablet personal computer, wherein the electronic device 20 includes an image capturing apparatus 21. The image capturing apparatus 21 includes an imaging system (its reference numeral is omitted) according to the present disclosure and an image sensor (its reference numeral is omitted), wherein the image sensor is disposed on an image surface of the imaging system.

13th Embodiment

[0213] FIG. 23 is a schematic view of an electronic device 30 according to the 13th embodiment of the present disclosure. The electronic device 30 of the 13th embodiment is a head-mounted display, wherein the electronic device 30 includes an image capturing apparatus 31. The image capturing apparatus 31 includes an imaging system (its reference numeral is omitted) according to the present disclosure and an image sensor (its reference numeral is omitted), wherein the image sensor is disposed on an image surface of the imaging system.

[0214] The foregoing description, for purpose of explanation, has been described with reference to specific embodiments. It is to be noted that TABLES 1-20 show different data of the different embodiments; however, the data of the different embodiments are obtained from experiments. The embodiments were chosen and described in order to best explain the principles of the disclosure and its practical appli-

cations, to thereby enable others skilled in the art to best utilize the disclosure and various embodiments with various modifications as are suited to the particular use contemplated. The embodiments depicted above and the appended drawings are exemplary and are not intended to be exhaustive or to limit the scope of the present disclosure to the precise forms disclosed. Many modifications and variations are possible in view of the above teachings.

What is claimed is:

1. An imaging system comprising, in order from an object side to an image side:

a first lens element with positive refractive power having an object-side surface being convex in a paraxial region thereof;

a second lens element with negative refractive power having an image-side surface being concave in a paraxial region thereof;

a third lens element having refractive power;

a fourth lens element with refractive power having an image-side surface being concave in a paraxial region thereof, wherein an object-side surface and the image-side surface of the fourth lens element are aspheric, and the image-side surface of the fourth lens element comprises at least one convex shape in an off-axial region thereof; and

a fifth lens element with refractive power having an object-side surface being convex in a paraxial region thereof and an image-side surface being concave in a paraxial region thereof, wherein the object-side surface and the image-side surface of the fifth lens element are aspheric, and the image-side surface of the fifth lens element comprises at least one convex shape in an off-axial region thereof;

wherein the imaging system has a total of five lens elements with refractive power, there is an air gap between any two of the first lens element, the second lens element, the third lens element, the fourth lens element and the fifth lens element that are adjacent to each other, a central thickness of the fourth lens element is CT4, an axial distance between the third lens element and the fourth lens element is T34, an axial distance between the fourth lens element and the fifth lens element is T45, a focal length of the imaging system is f, a curvature radius of an object-side surface of the third lens element is R5, a curvature radius of an image-side surface of the third lens element is R6, an Abbe number of the second lens element is V2, an Abbe number of the third lens element is V3, an Abbe number of the fourth lens element is V4, and the following conditions are satisfied:

$$CT4/T34 < 1.20;$$

$$CT4/T45 < 1.60;$$

$$|f/R5| + |f/R6| < 1.10; \text{ and}$$

$$1.75 < (V2 + V4)/V3.$$

2. The imaging system of claim 1, wherein the object-side surface of the fourth lens element is convex in a paraxial region thereof.

3. The imaging system of claim 2, wherein the Abbe number of the third lens element is V3, and the following condition is satisfied:

$$V3 < 30.$$

4. The imaging system of claim 2, wherein the central thickness of the fourth lens element is CT4, the axial distance between the third lens element and the fourth lens element is T34, and the following condition is satisfied:

$$CT4/T34 < 1.0.$$

5. The imaging system of claim 1, wherein the central thickness of the fourth lens element is CT4, the axial distance between the fourth lens element and the fifth lens element is T45, and the following condition is satisfied:

$$CT4/T45 < 1.48.$$

6. The imaging system of claim 1, wherein the central thickness of the fourth lens element is CT4, the axial distance between the fourth lens element and the fifth lens element is T45, and the following condition is satisfied:

$$CT4/T45 < 1.25.$$

7. The imaging system of claim 1, wherein the third lens element has positive refractive power.

8. The imaging system of claim 1, wherein the first lens element has an image-side surface being concave in a paraxial region thereof.

9. The imaging system of claim 1, wherein a curvature radius of the object-side surface of the fourth lens element is R7, the focal length of the imaging system is f, and the following condition is satisfied:

$$|R7/f| < 3.0.$$

10. The imaging system of claim 9, wherein an f-number of the imaging system is Fno, an axial distance between the object-side surface of the first lens element and an image surface is TL, a half of a maximal field of view of the imaging system is HFOV, and the following conditions are satisfied:

$$1.6 < Fno < 2.4;$$

$$TL < 6 \text{ mm}; \text{ and}$$

$$35 \text{ degrees} < HFOV.$$

11. The imaging system of claim 1, wherein a curvature radius of the object-side surface of the fourth lens element is R7, the focal length of the imaging system is f, and the following condition is satisfied:

$$|R7/f| < 2.3.$$

12. The imaging system of claim 1, wherein the focal length of the imaging system is f, a focal length of the third lens element is f3, a focal length of the fourth lens element is f4, a focal length of the fifth lens element is f5, and the following condition is satisfied:

$$|f/f3| + |f/f4| + |f/f5| < 0.60.$$

13. The imaging system of claim 1, wherein a focal length of the first lens element is f1, a focal length of the second lens element is f2, a focal length of the third lens element is f3, a focal length of the fourth lens element is f4, a focal length of the fifth lens element is f5, and the following conditions are satisfied:

$$|f3| > |f1|;$$

$$|f3| > |f2|;$$

$$|f3| > |f4|; \text{ and}$$

$$|f3| > |f5|.$$

14. The imaging system of claim 1, wherein a curvature radius of the object-side surface of the fourth lens element is R7, a curvature radius of the image-side surface of the fourth lens element is R8, a curvature radius of the object-side surface of the fifth lens element is R9, a curvature radius of the image-side surface of the fifth lens element is R10, the focal length of the imaging system is f, and the following condition is satisfied:

$$(|R7| + |R8| + |R9| + |R10|)/f < 3.5.$$

15. The imaging system of claim 1, wherein the axial distance between the third lens element and the fourth lens element is T34, the axial distance between the fourth lens element and the fifth lens element is T45, a sum of axial distances between any two of the first lens element, the second lens element, the third lens element, the fourth lens element and the fifth lens element that are adjacent to each other is ΣAT, and the following condition is satisfied:

$$0.60 < (T34 + T45)/\Sigma AT < 0.80.$$

16. The imaging system of claim 1, wherein the Abbe number of the second lens element is V2, the Abbe number of the third lens element is V3, the Abbe number of the fourth lens element is V4, and the following condition is satisfied:

$$V2 + V3 + V4 < 90.$$

17. The imaging system of claim 1, wherein a central thickness of the first lens element is CT1, a central thickness of the second lens element is CT2, a central thickness of the third lens element is CT3, the central thickness of the fourth lens element is CT4, a central thickness of the fifth lens element is CT5, and CT5 is larger than CT1, CT2, CT3, and CT4.

18. The imaging system of claim 1, wherein a central thickness of the second lens element is CT2, a central thickness of the third lens element is CT3, the central thickness of the fourth lens element is CT4, a central thickness of the fifth lens element is CT5, and the following condition is satisfied:

$$(CT2 + CT3 + CT4)/CT5 < 1.25.$$

19. The imaging system of claim 1, wherein a curvature radius of the image-side surface of the fourth lens element is R8, a curvature radius of the object-side surface of the fifth lens element is R9, and the following condition is satisfied:

$$1.20 < R8/R9.$$

20. The imaging system of claim 1, wherein the focal length of the imaging system is f, the curvature radius of the object-side surface of the third lens element is R5, the curvature radius of the image-side surface of the third lens element is R6, and the following condition is satisfied:

$$|f/R5| + |f/R6| < 0.75.$$

21. An image capturing apparatus, comprising:

the imaging system of claim 1; and

an image sensor, wherein the image sensor is disposed on an image surface of the imaging system.

22. An electronic device, comprising:

the image capturing apparatus of claim 21.

* * * * *